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(54) **DIMENSION CREATION DEVICE,
DIMENSION CREATION METHOD, AND
RECORDING MEDIUM**

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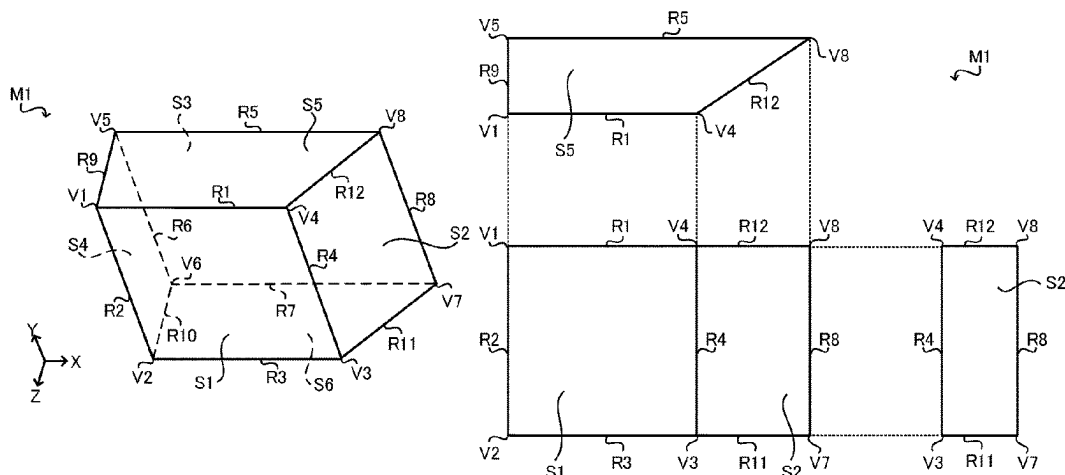
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CPC **G06F 30/10** (2020.01)

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CPC G06F 30/10
See application file for complete search history.

(57) **ABSTRACT**

In a dimension creation device, an element selector selects a first target element and a second target element from elements of a three-dimensional model. A direction indicator indicates a direction of a dimension to be created. A dimension creator creates, when the element selector selects a vertex or an edge as each of the first target element and the second target element and the three-dimensional model includes a first orthogonal face including the first target element and orthogonal to the direction indicated by the direction indicator and a second orthogonal face including the second target element and orthogonal to the direction indicated by the direction indicator, the dimension between the first orthogonal face and the second orthogonal face in the direction indicated by the direction indicator in a manner associated with the first orthogonal face and the second orthogonal face.

13 Claims, 17 Drawing Sheets



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FIG. 1

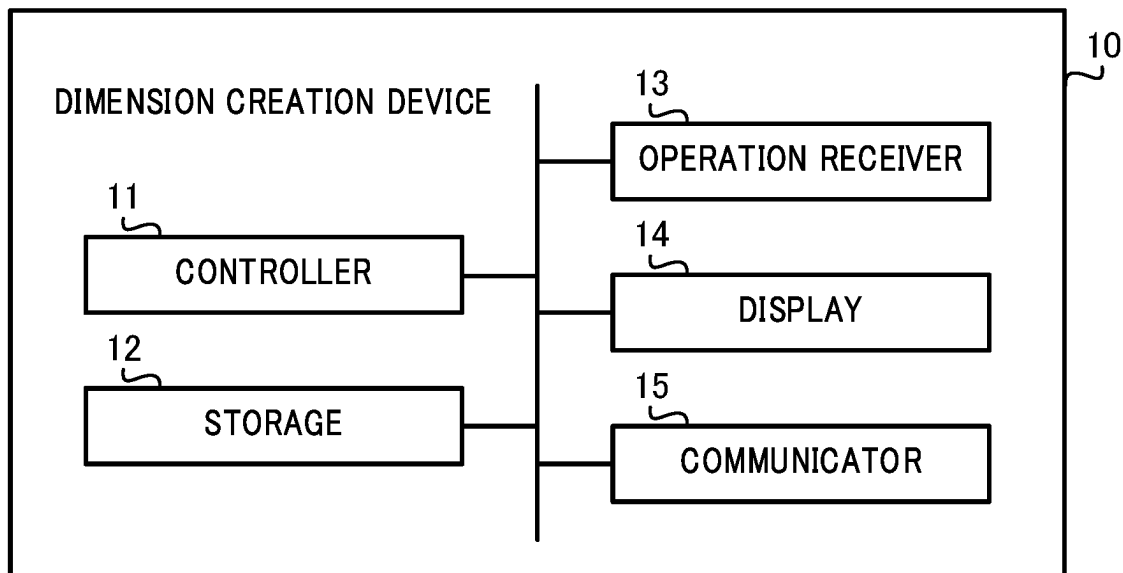


FIG. 2

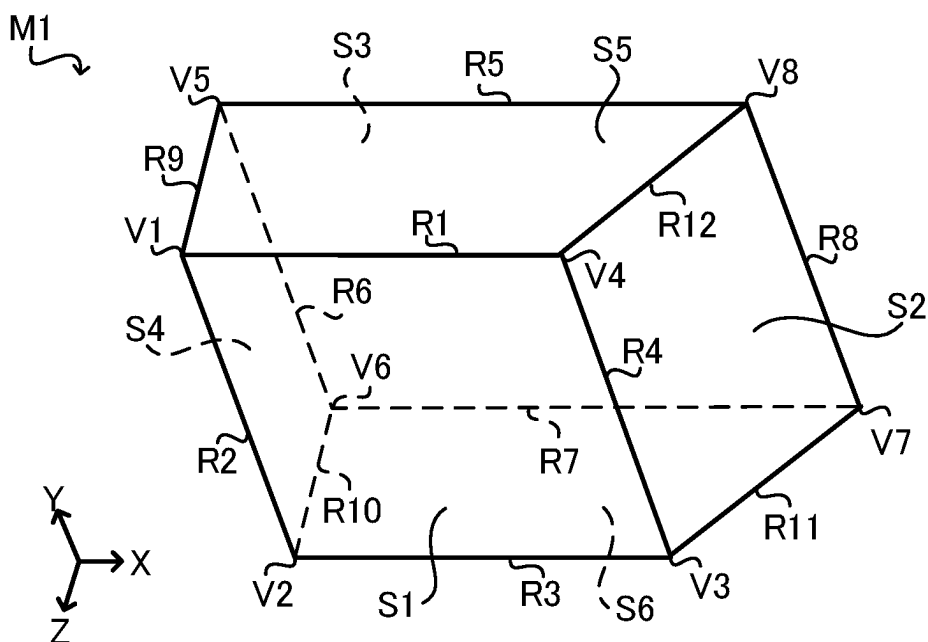


FIG.3

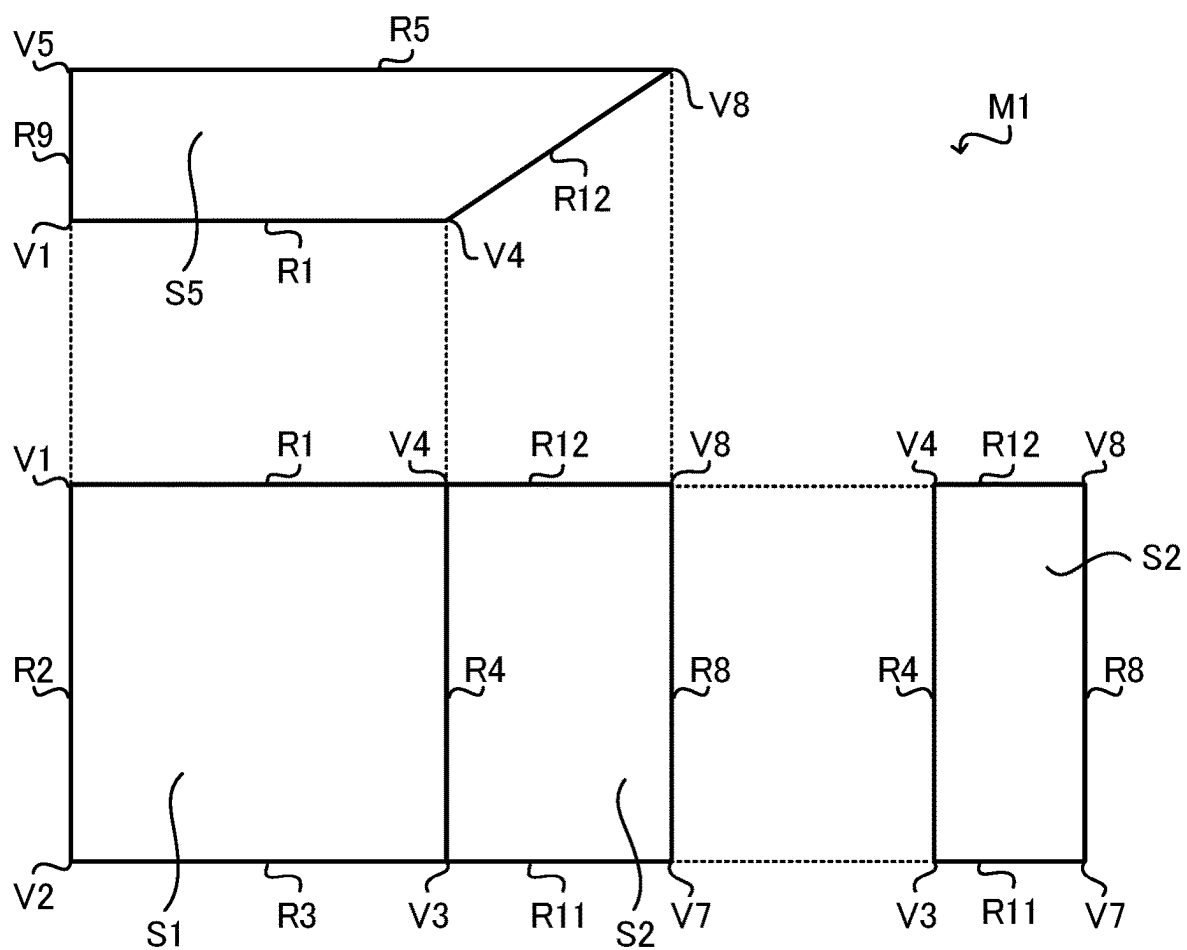


FIG.4

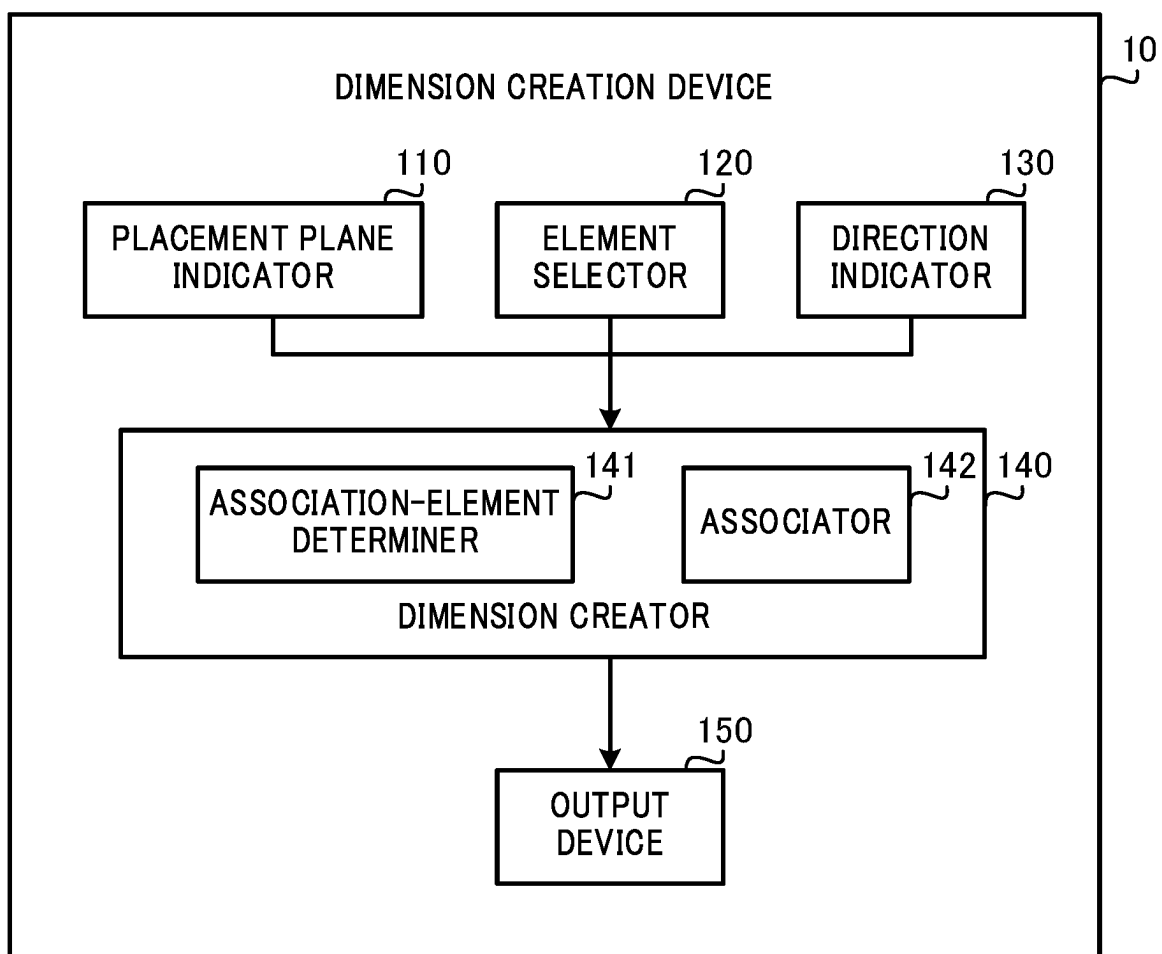


FIG.5

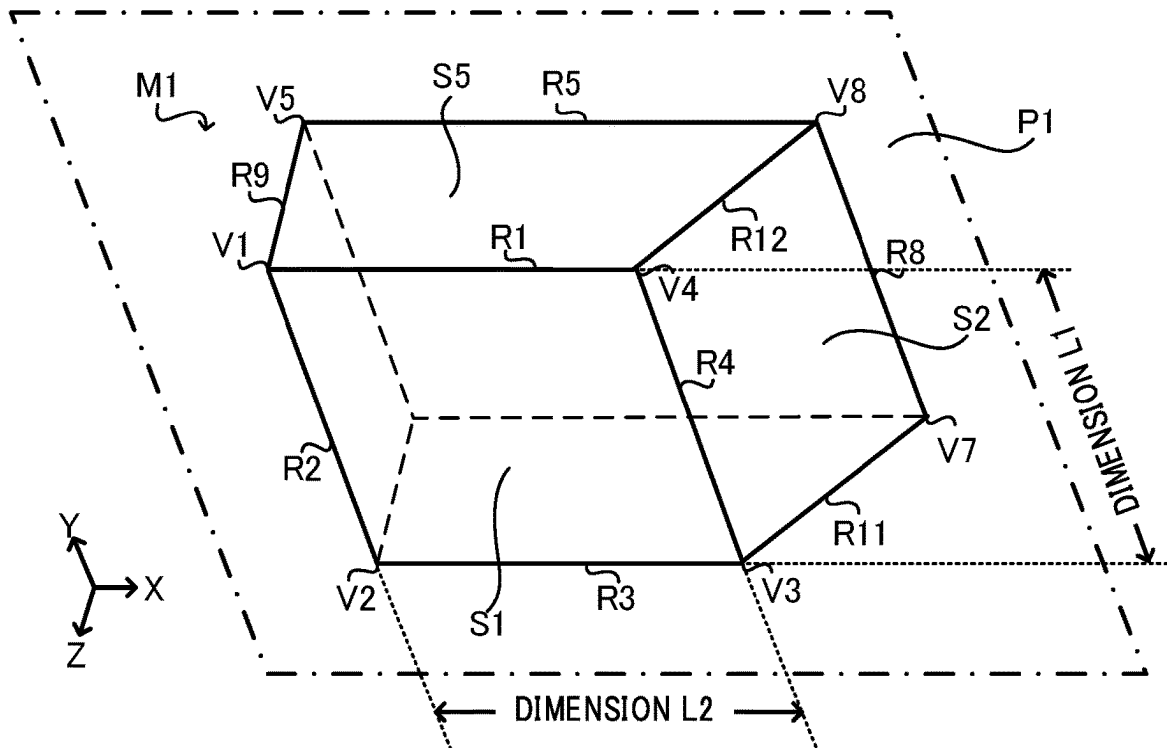


FIG.6

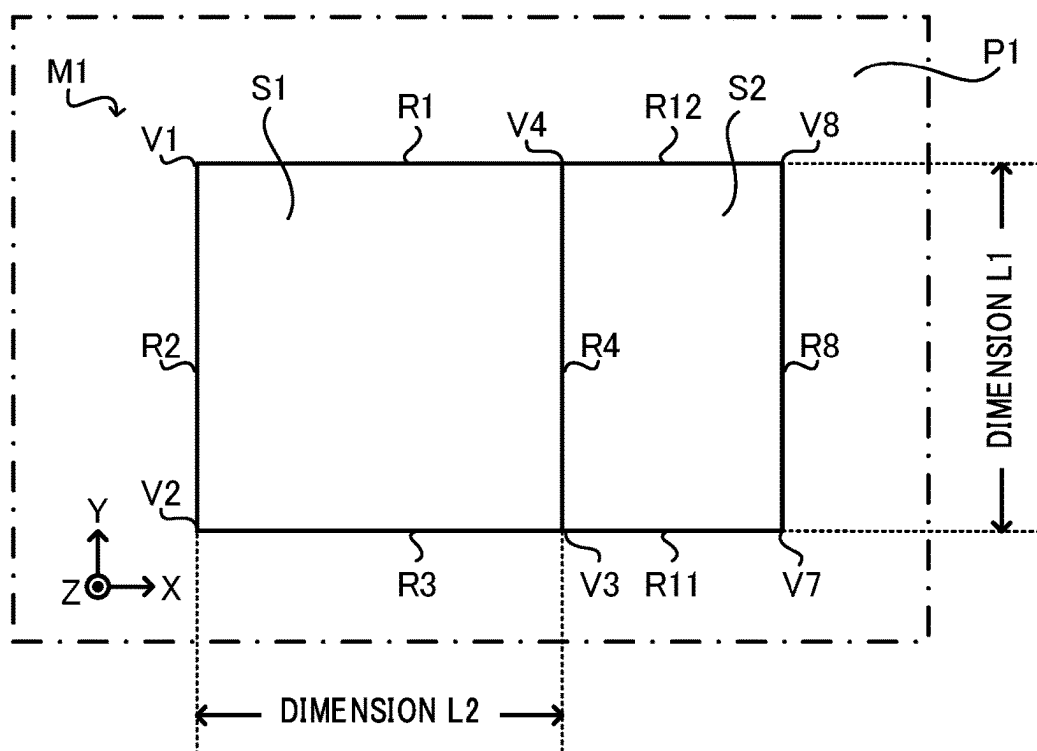


FIG. 7

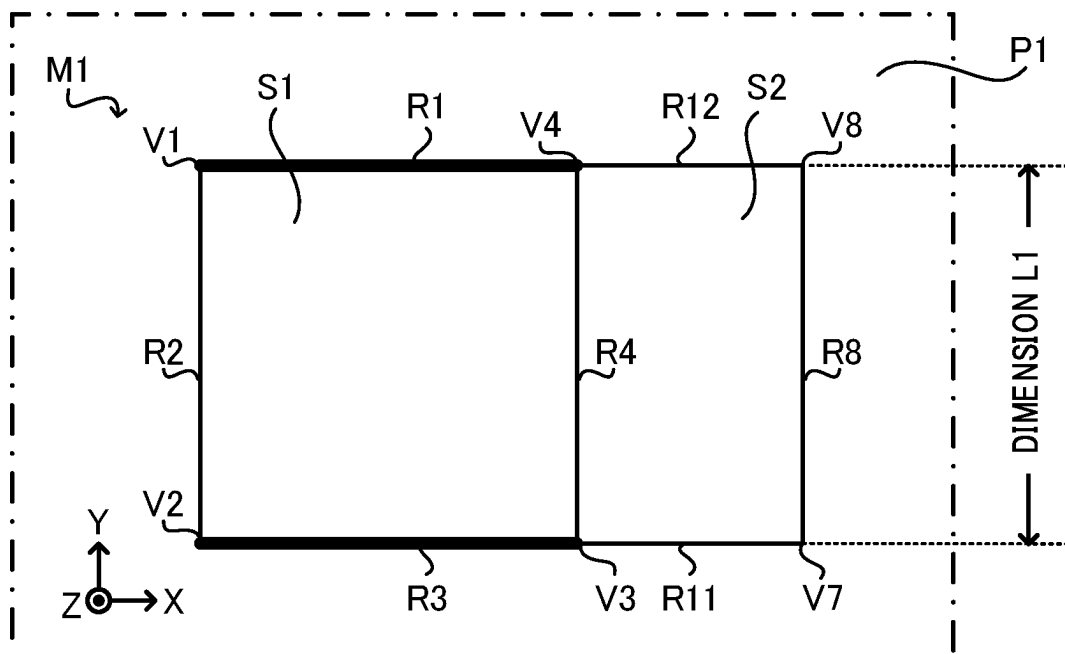


FIG. 8

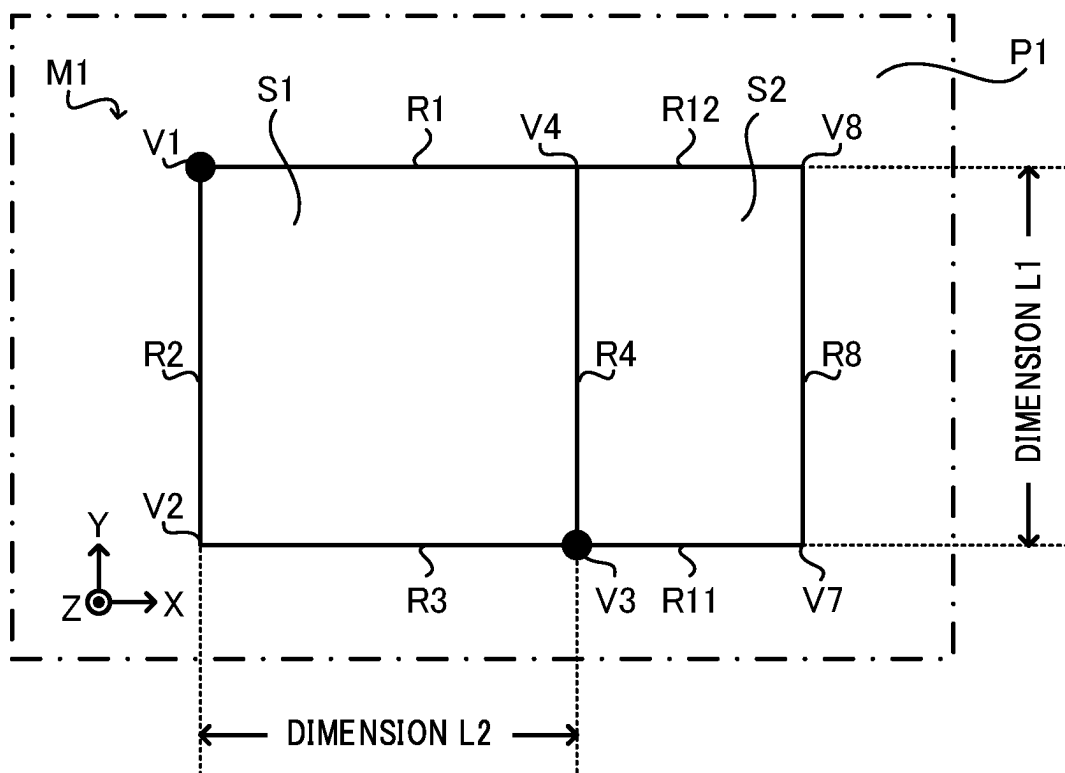


FIG.9

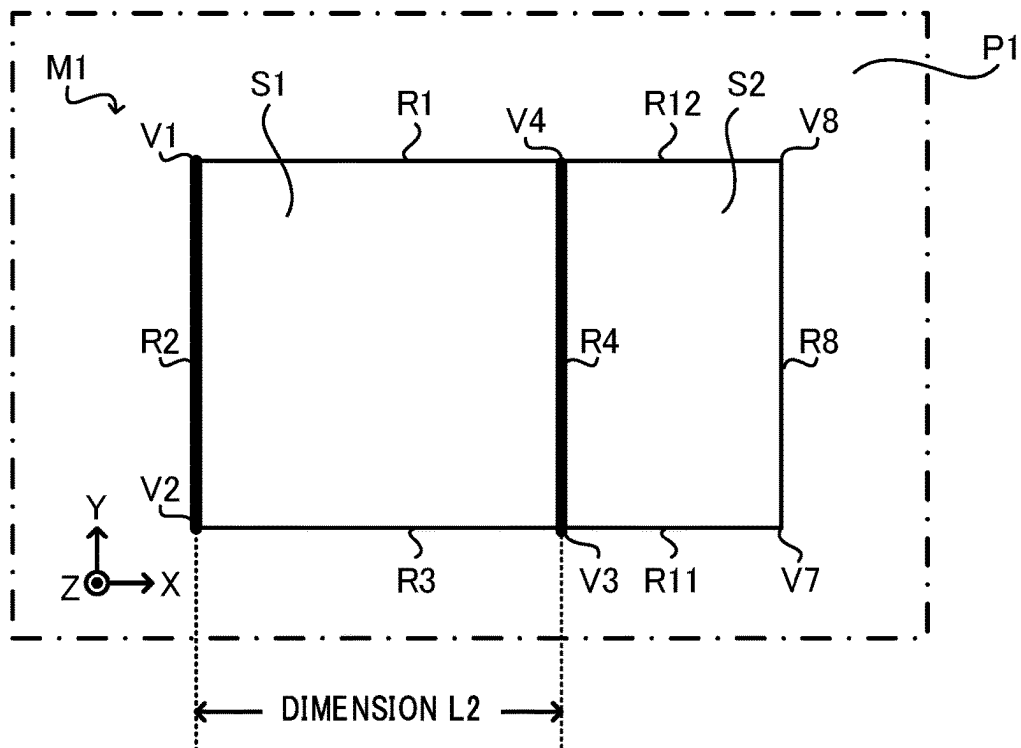


FIG.10

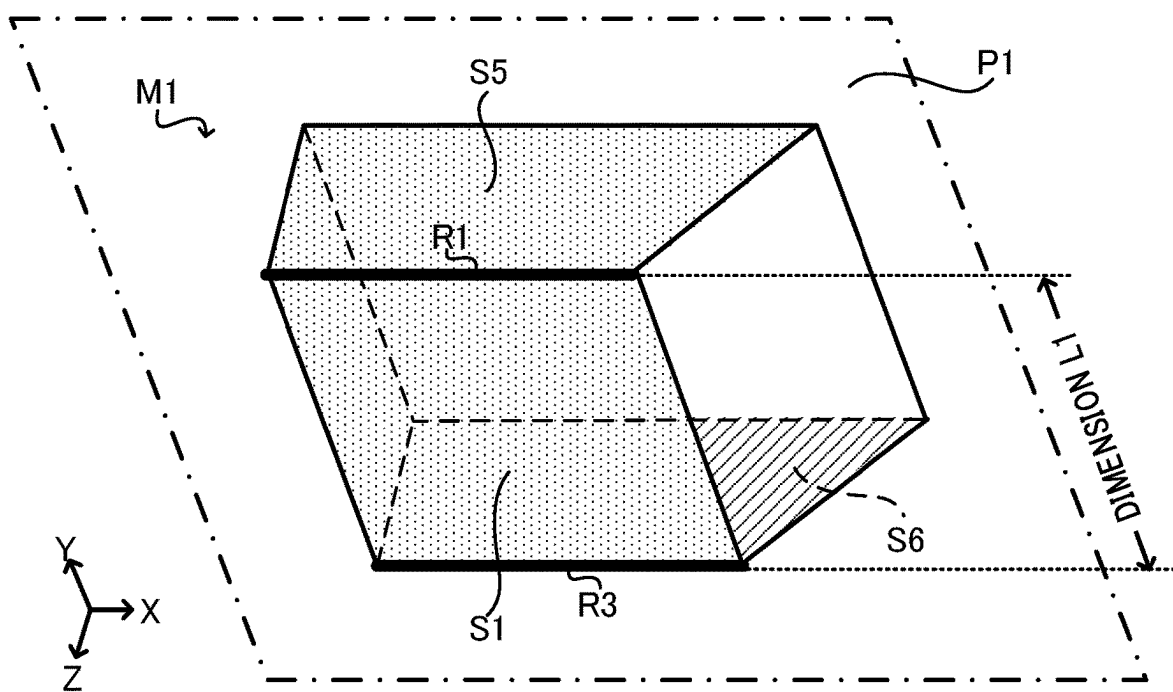


FIG.11

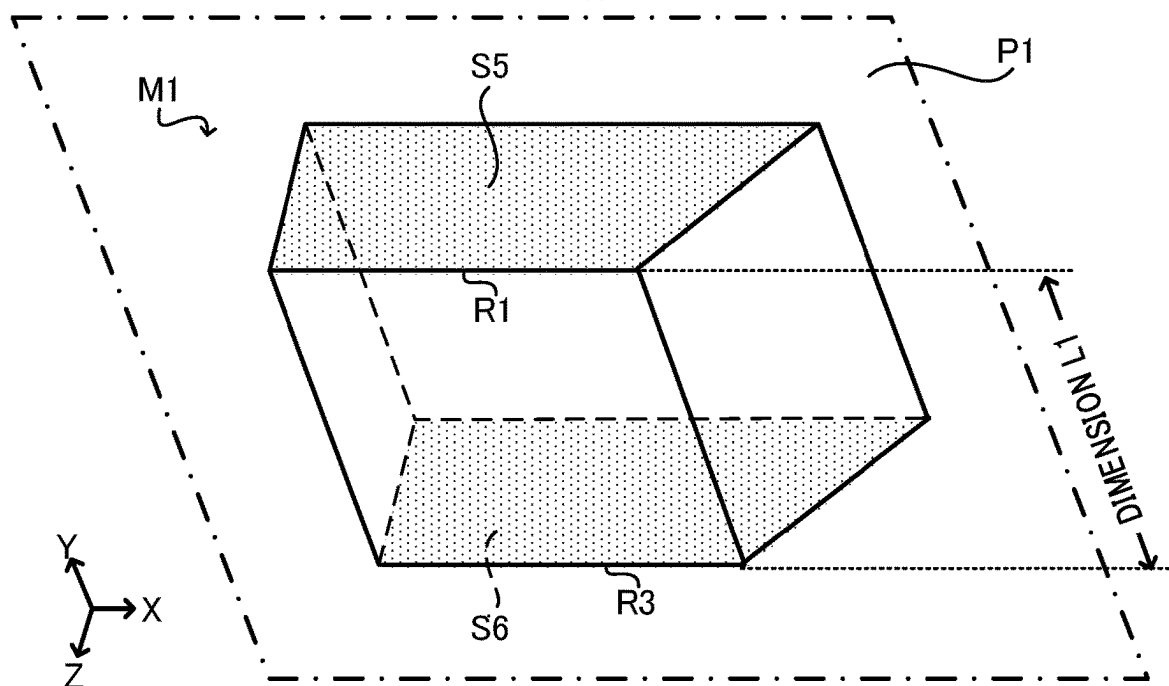


FIG.12

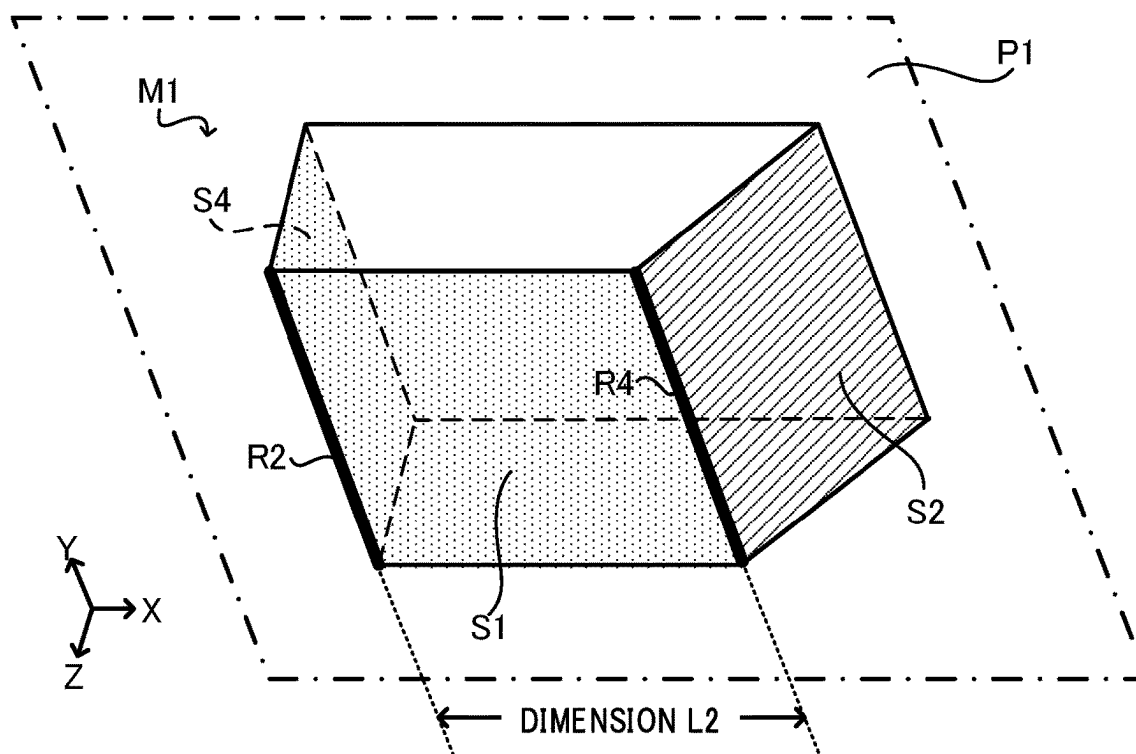


FIG.13

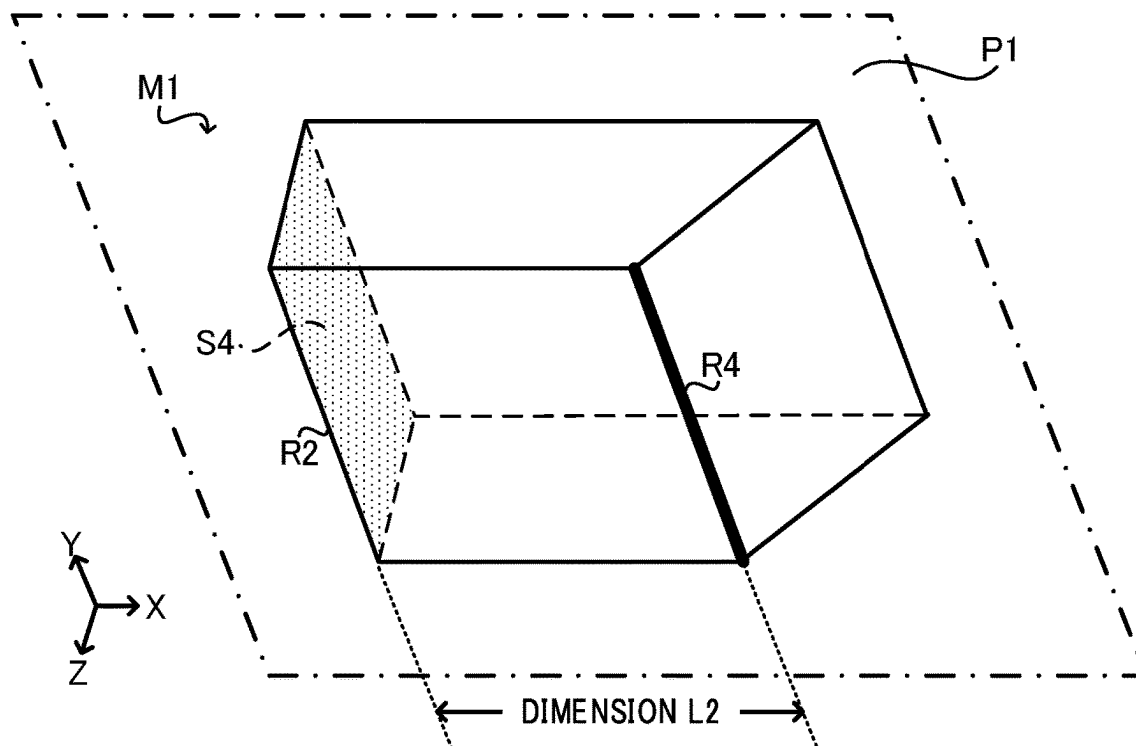


FIG.14

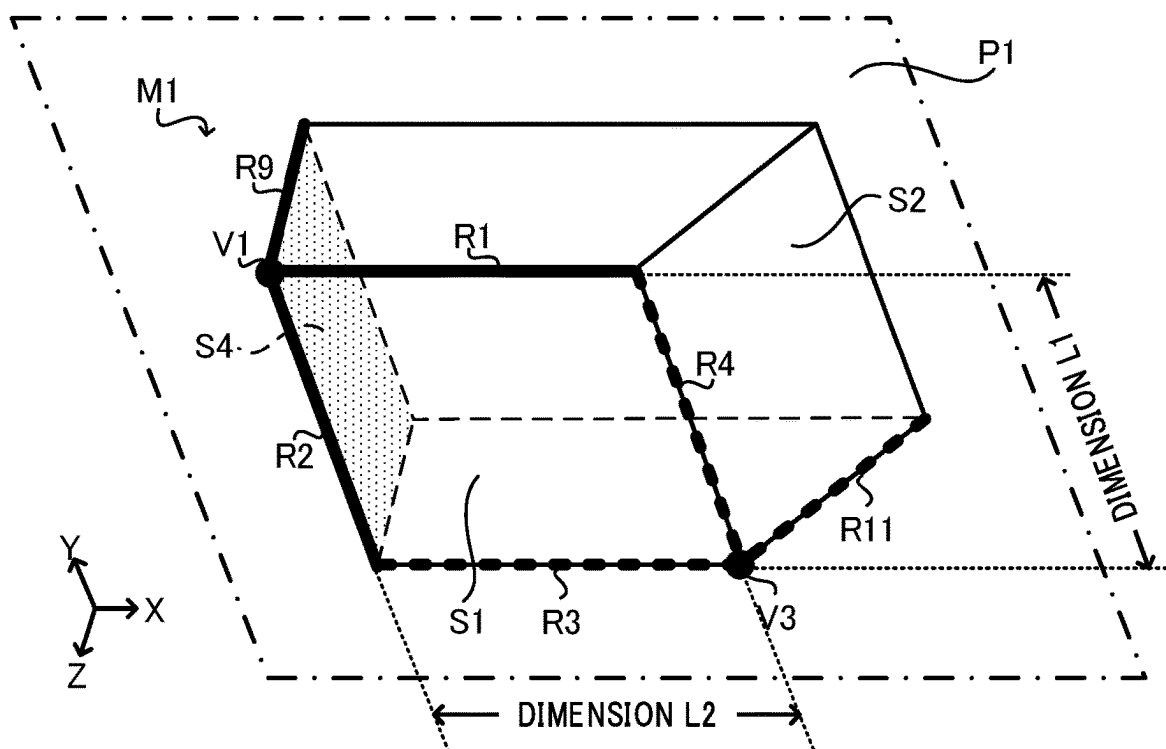


FIG.15

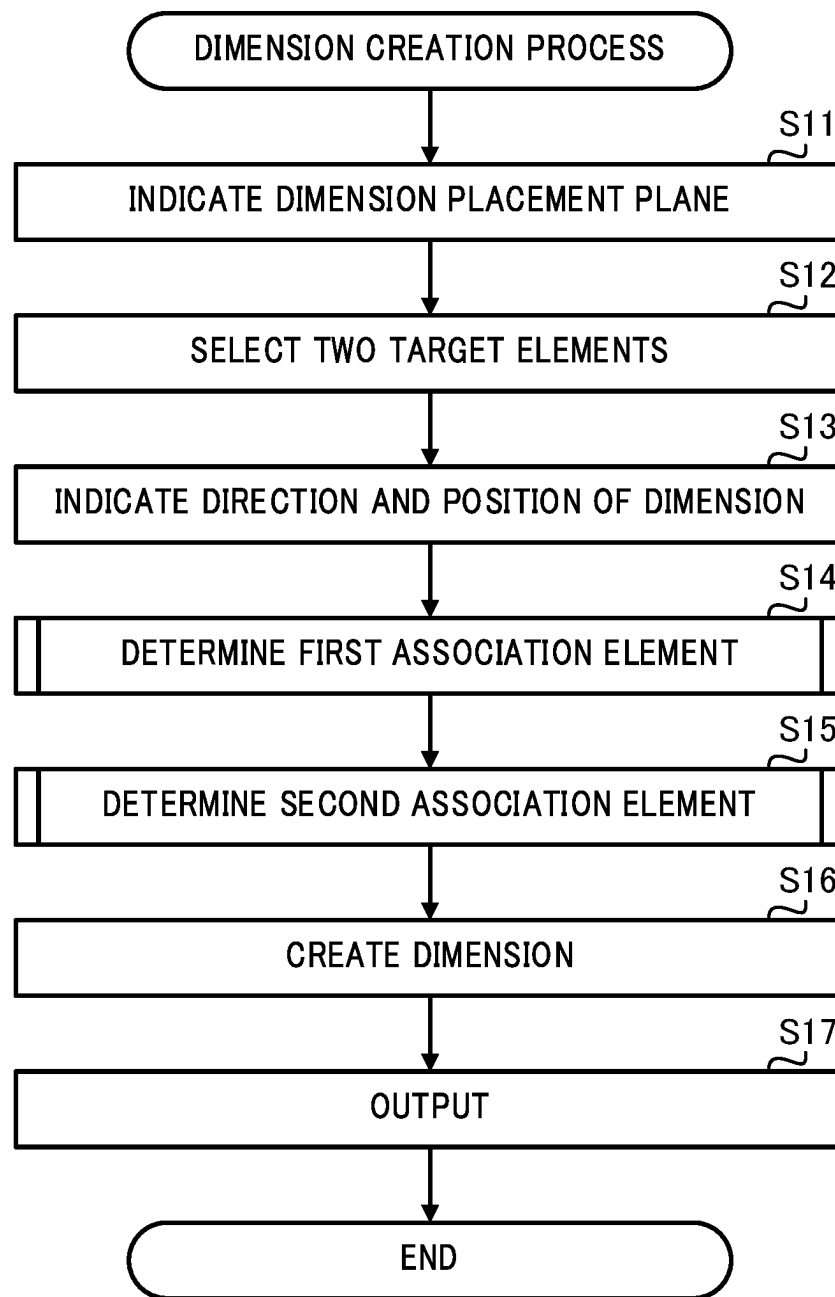


FIG.16

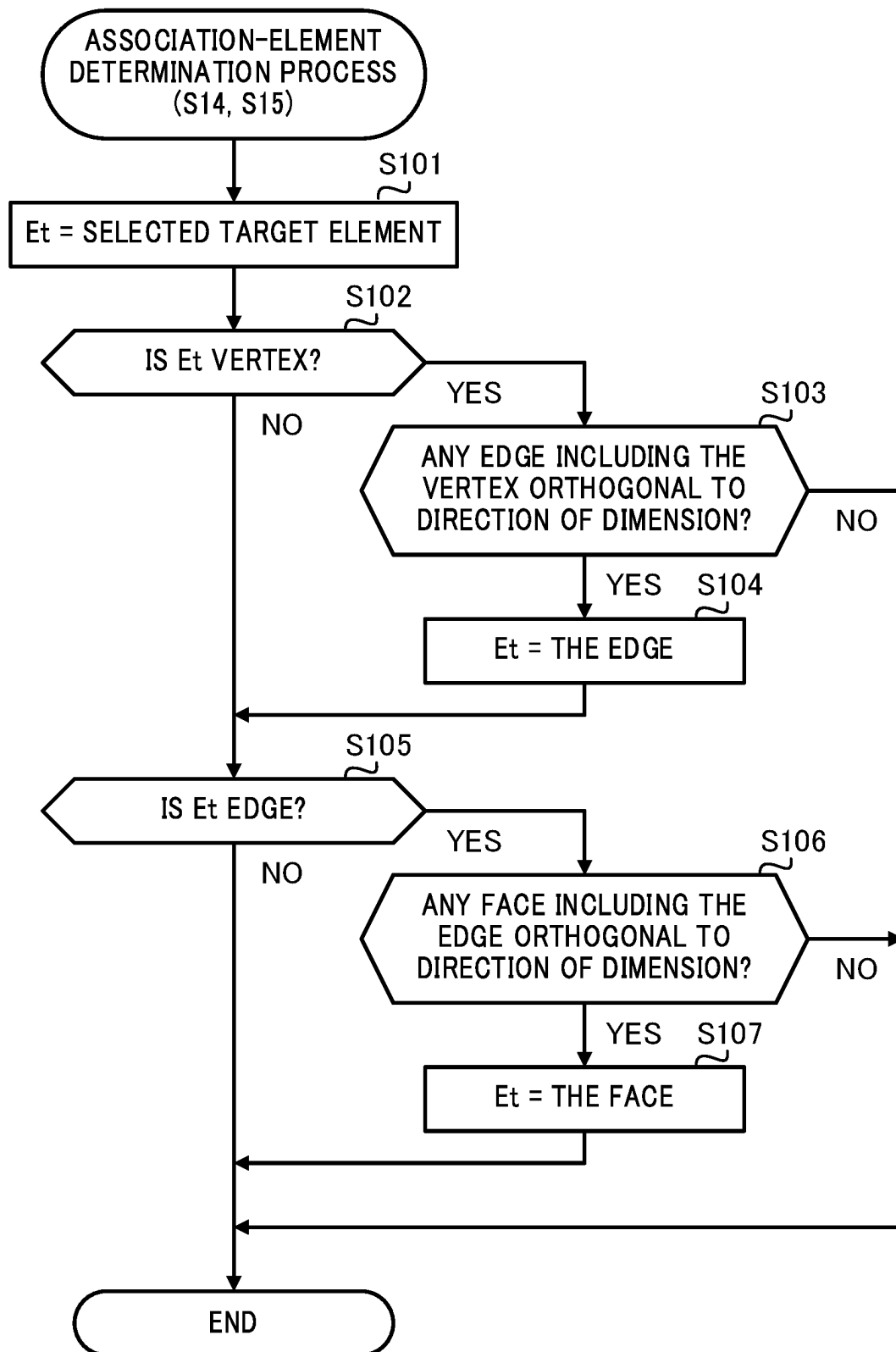


FIG.17

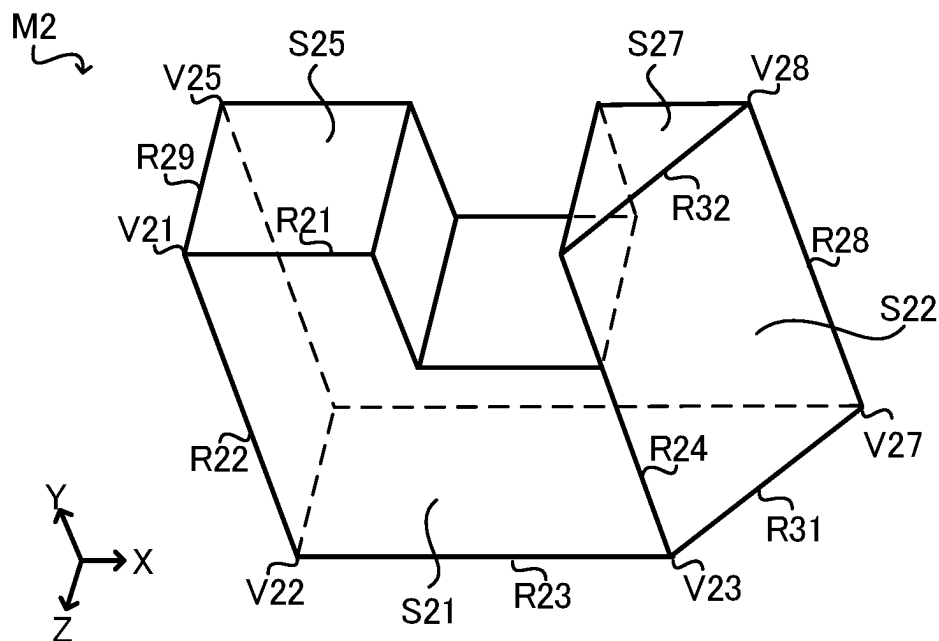


FIG.18

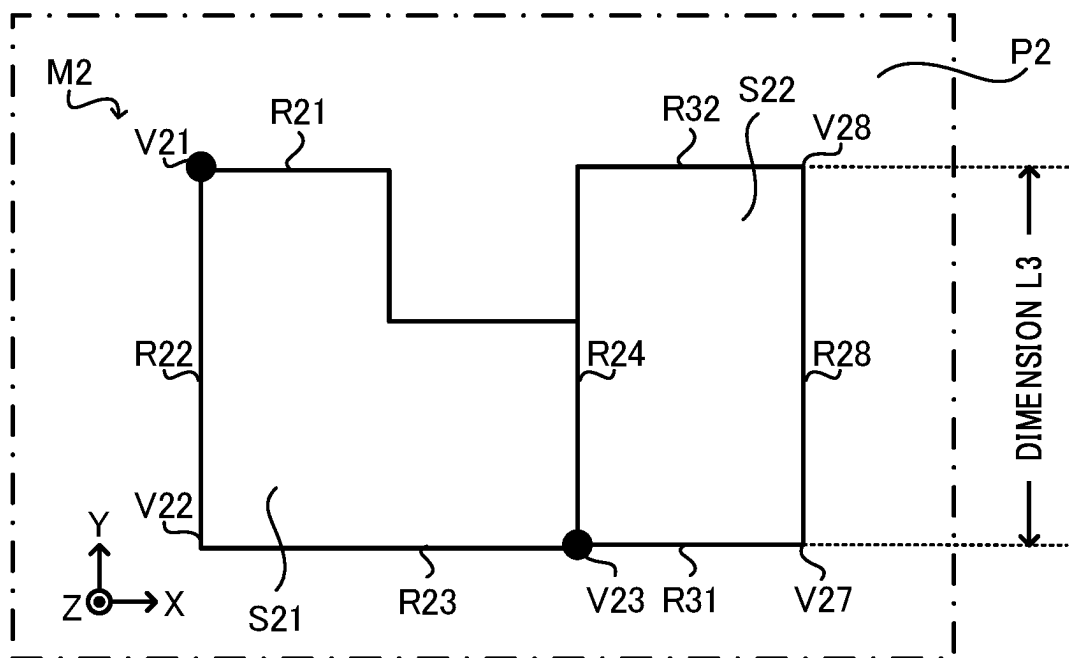


FIG.19

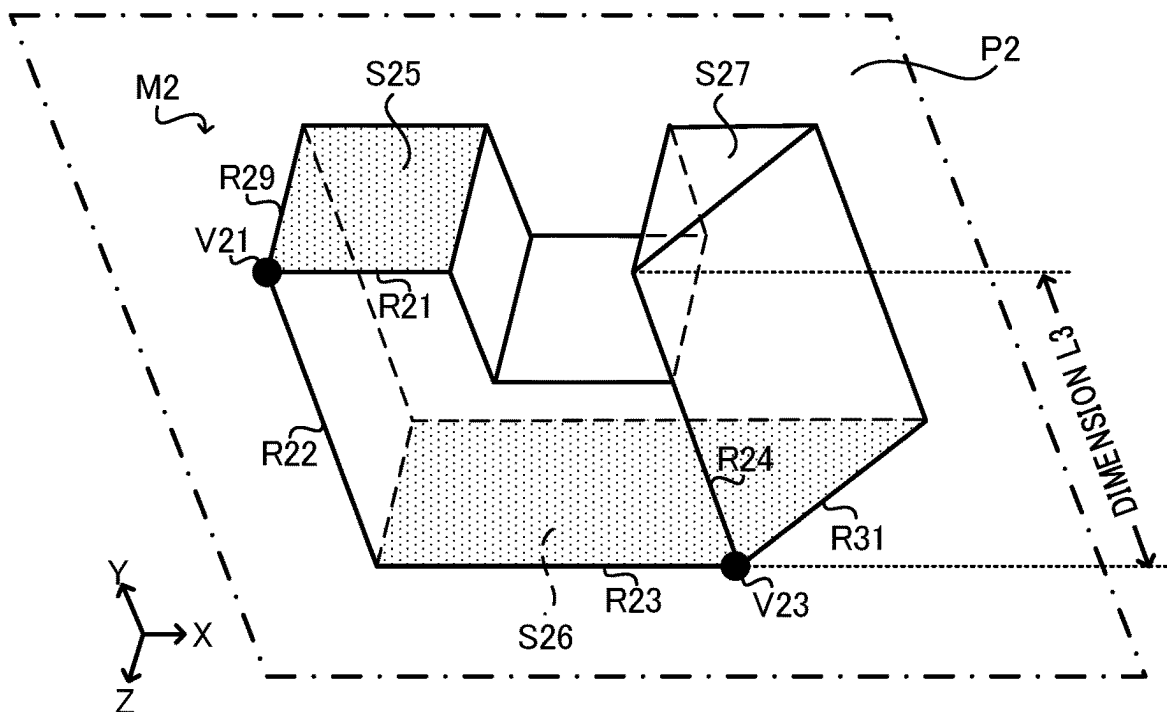


FIG.20

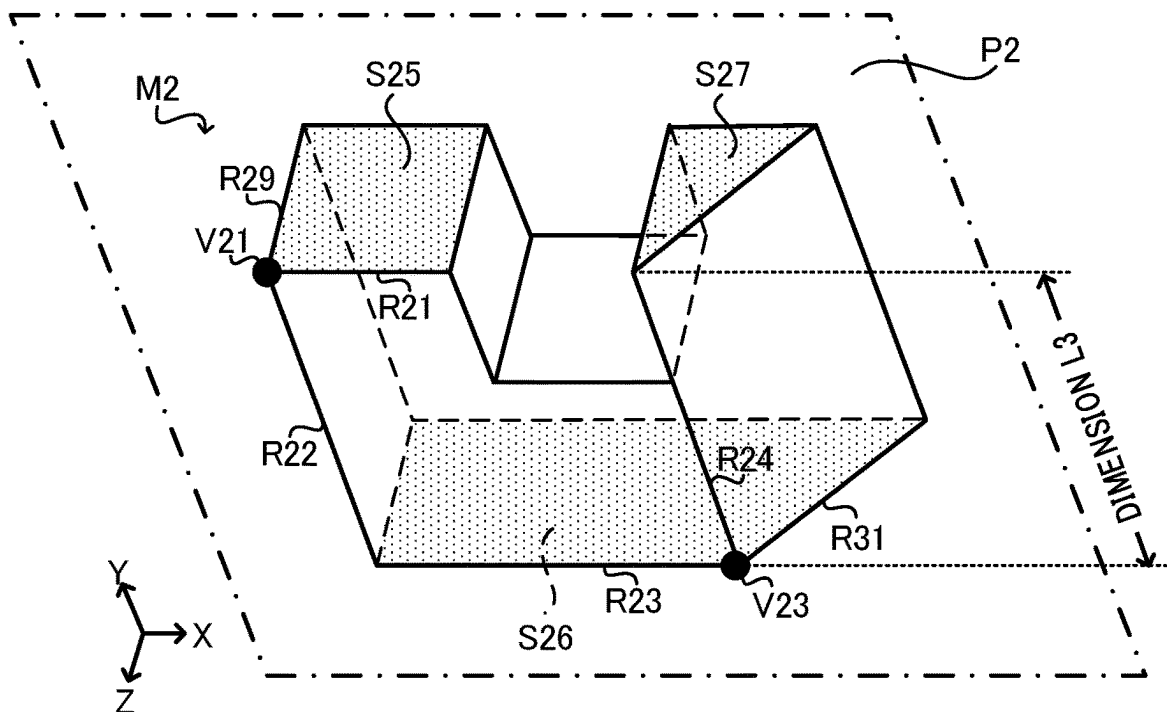


FIG.21

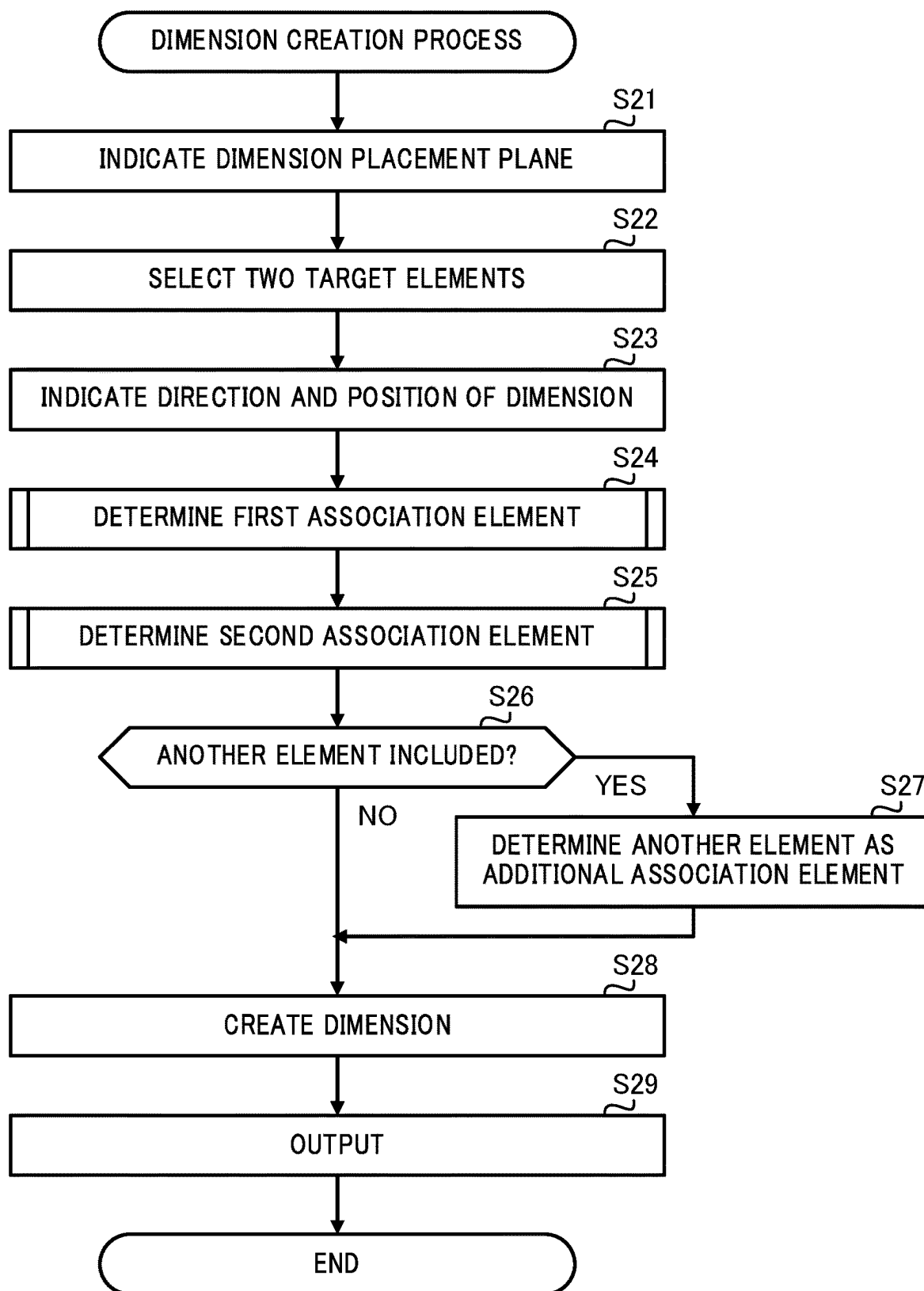


FIG.22

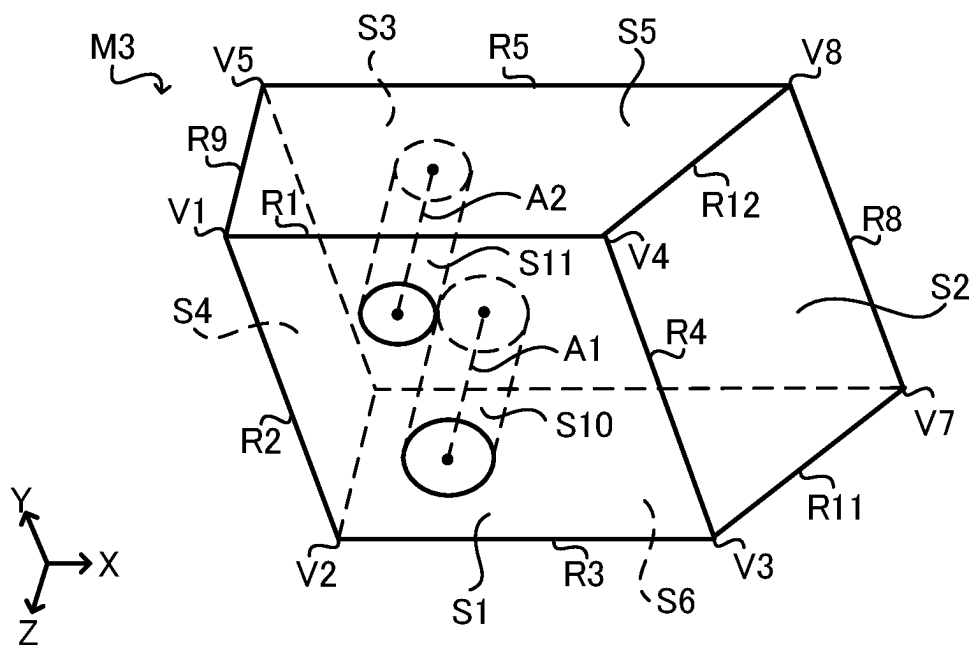


FIG.23

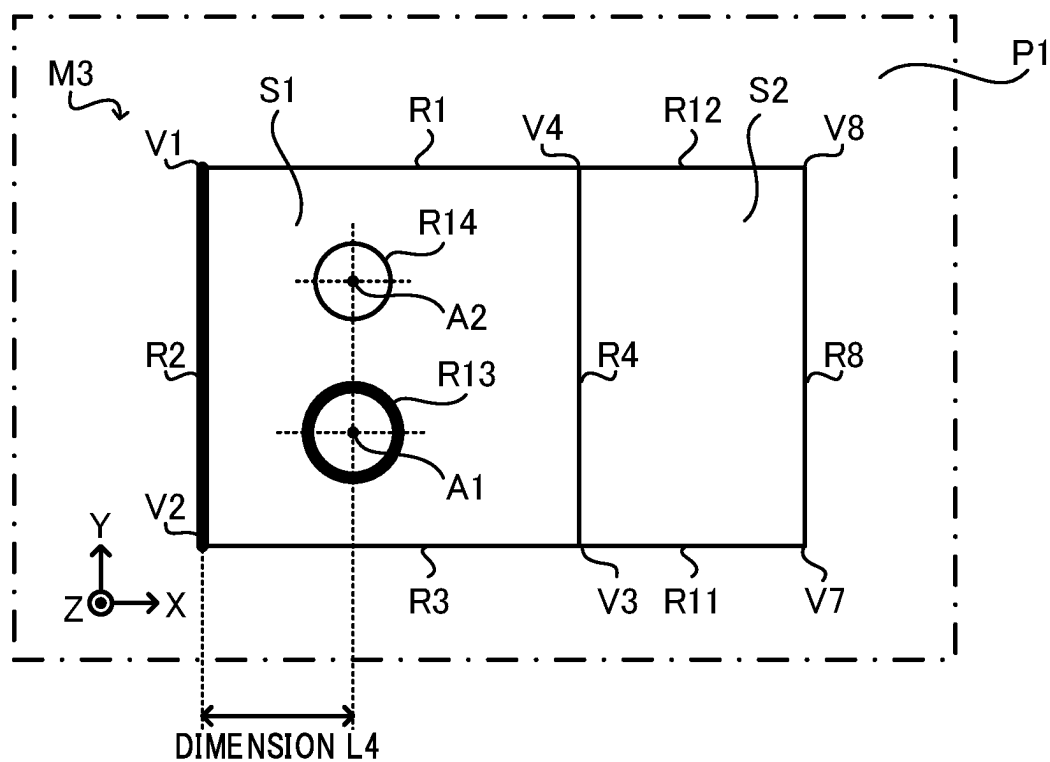


FIG.24

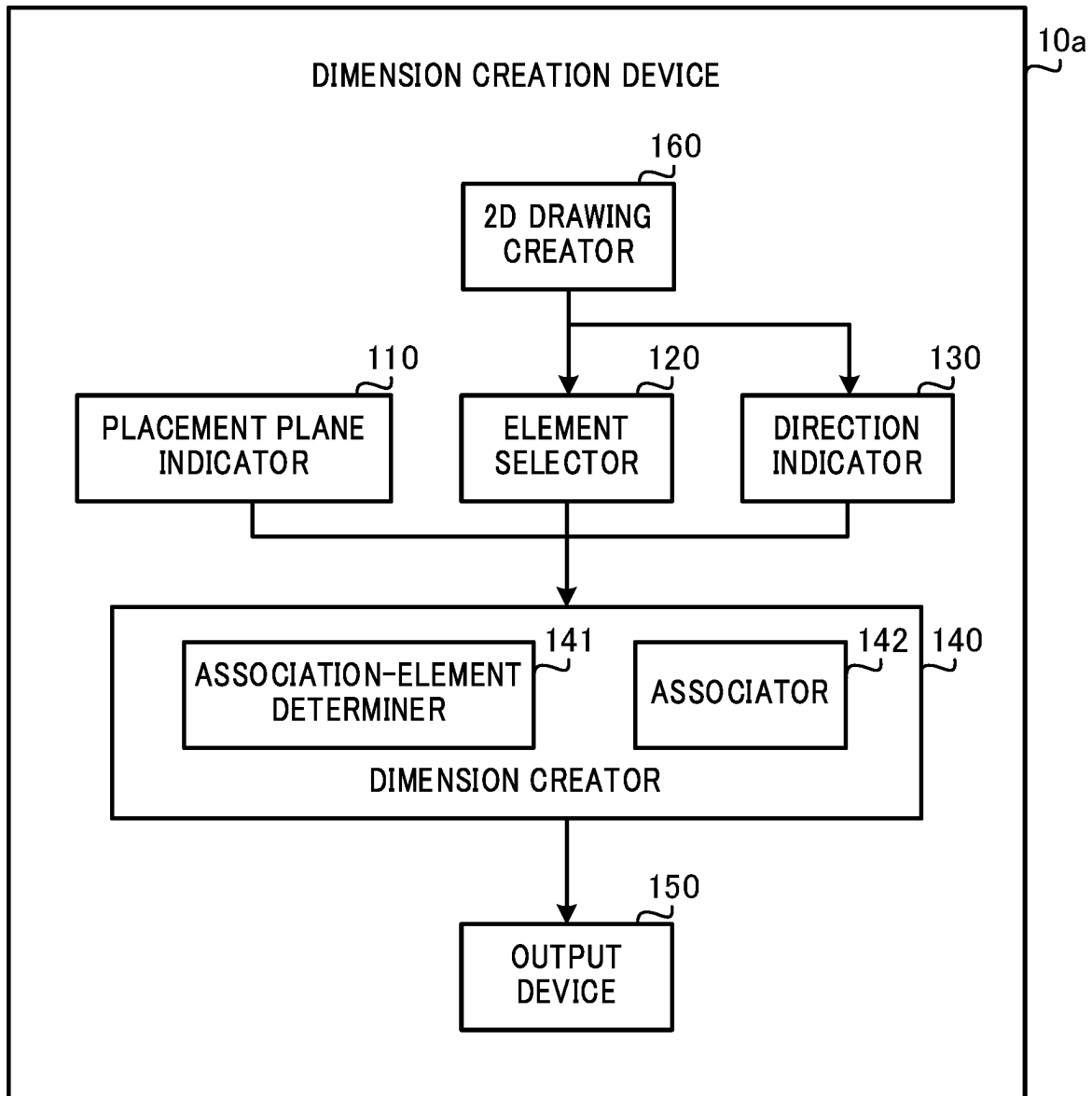


FIG.25

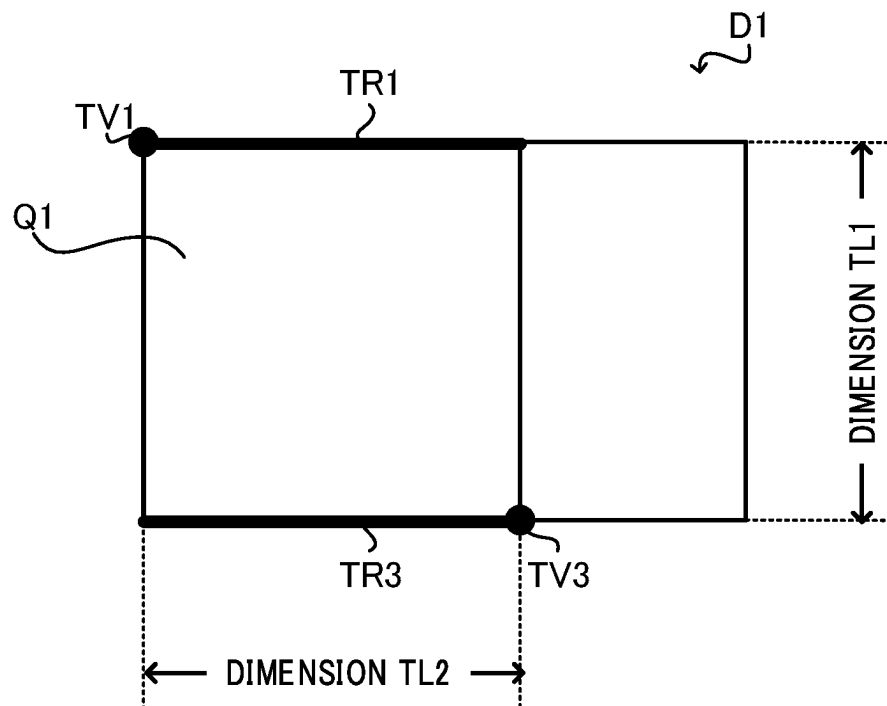
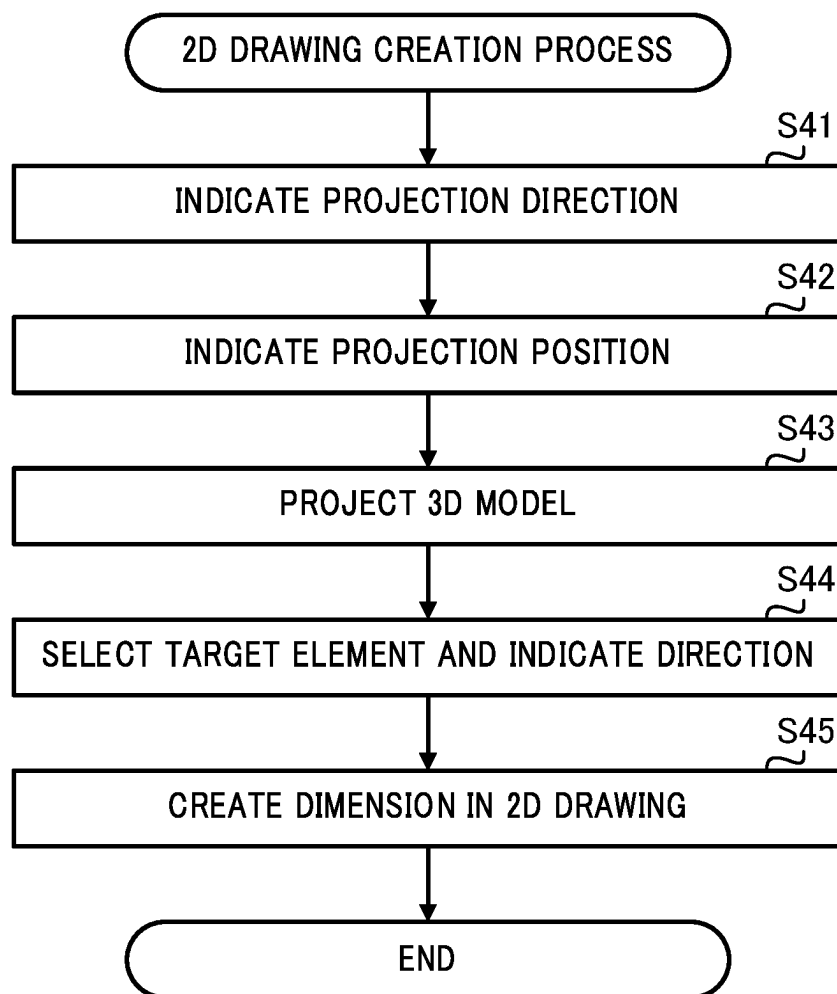


FIG.26



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DIMENSION CREATION DEVICE, DIMENSION CREATION METHOD, AND RECORDING MEDIUM

TECHNICAL FIELD

The present disclosure relates to a dimension creation device, a dimension creation method, and a program.

BACKGROUND ART

A technique is known for creating dimensions for a three-dimensional (3D) model and annotating the model with the dimensions. For example, Patent Literature 1 describes a face selection method for creating attribute information including dimensions for a 3D model displayed on a screen with a 3D computer aided design (CAD) device. More specifically, the face selection method described in Patent Literature 1 includes selecting, when an edge of a 3D model is selected, one of the faces forming the selected edge and not displayed on the screen to simplify the procedure of selecting an intended face and reduce the burden on an operator.

CITATION LIST

Patent Literature

Patent Literature 1: Unexamined Japanese Patent Application Publication No. 2006-72543

SUMMARY OF INVENTION

Technical Problem

The dimensions created for a 3D model are to be correctly associated with elements of the 3D model. However, the elements of a 3D model include faces, edges, and vertices. The elements of a 3D model are selected typically by moving or rotating a 3D model displayed on a screen to change the displayed state. While moving or rotating the 3D model displayed on the screen, the operator is to correctly determine and select faces, edges, or vertices as elements to be associated with the dimension. Thus, creating the dimension in a manner correctly associated with the elements of the 3D model takes more time and effort than creating the dimension in a two-dimensional (2D) drawing.

In response to the above issue, an objective of the present disclosure is to provide a dimension creation device and others that can efficiently create a dimension correctly associated with elements of a 3D model.

Solution to Problem

To achieve the above objective, a dimension creation device according to an aspect of the present disclosure includes element selection means for selecting a first target element and a second target element from elements of a three-dimensional model, direction indication means for indicating a direction of a dimension to be created, and dimension creation means for creating, when the element selection means selects a vertex or an edge as each of the first target element and the second target element and the three-dimensional model includes (i) a first orthogonal face including the first target element and orthogonal to the direction indicated by the direction indication means and (ii) a second orthogonal face including the second target ele-

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ment and orthogonal to the direction indicated by the direction indication means, the dimension between the first orthogonal face and the second orthogonal face in the direction indicated by the direction indication means in a manner associated with the first orthogonal face and the second orthogonal face.

Advantageous Effects of Invention

The structure according to the above aspect of the present disclosure selects a first target element and a second target element from elements of a 3D model, indicates a direction of a dimension to be created, and creates, when a vertex or an edge is selected as each of the first target element and the second target element and the 3D model includes a first orthogonal face including the first target element and orthogonal to the indicated direction and a second orthogonal face including the second target element and orthogonal to the indicated direction, the dimension between the first orthogonal face and the second orthogonal face in the indicated direction in a manner associated with the first orthogonal face and the second orthogonal face. Thus, the structure according to the above aspect of the present disclosure can efficiently create dimensions correctly associated with elements of a 3D model.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram of a dimension creation device according to Embodiment 1, showing the overall configuration;

FIG. 2 is a diagram of an example 3D model for which dimensions are created with a dimension creation device according to Embodiment 1;

FIG. 3 is a diagram of the 3D model shown in FIG. 2 in a 2D plane;

FIG. 4 is a functional block diagram of the dimension creation device according to Embodiment 1;

FIG. 5 is a diagram of the 3D model with an example dimension placement plane being indicated in Embodiment 1;

FIG. 6 is a diagram of the 3D model displayed with the direction of the dimension placement plane aligned with the direction of a screen in Embodiment 1;

FIG. 7 is a diagram of the 3D model with two edges of the 3D model being selected to create a dimension in the Y-direction in Embodiment 1;

FIG. 8 is a diagram of the 3D model with two vertices of the 3D model being selected to create a dimension in the X-direction or the Y-direction in Embodiment 1;

FIG. 9 is a diagram of the 3D model with two edges of the 3D model being selected to create a dimension in the X-direction in Embodiment 1;

FIG. 10 is a diagram of a first example face including the edges selected as target elements in Embodiment 1;

FIG. 11 is a diagram of example two faces determined as association elements in Embodiment 1;

FIG. 12 is a diagram of a second example face including edges selected as target elements in Embodiment 1;

FIG. 13 is a diagram of an example face and an example edge determined as association elements in Embodiment 1;

FIG. 14 is a diagram of example edges including vertices selected as target elements in Embodiment 1;

FIG. 15 is a flowchart of a dimension creation process performed by the dimension creation device according to Embodiment 1;

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FIG. 16 is a flowchart of an association-element determination process performed by the dimension creation device according to Embodiment 1;

FIG. 17 is a diagram of an example 3D model for which dimensions are created with a dimension creation device according to Embodiment 2;

FIG. 18 is a diagram of the 3D model with two vertices of the 3D model being selected to create a dimension in the Y-direction in Embodiment 2;

FIG. 19 is a diagram of example two faces determined as association elements in Embodiment 2;

FIG. 20 is a diagram of another example face determined as an additional association element in Embodiment 2, in addition to the two faces shown in FIG. 19;

FIG. 21 is a flowchart of a dimension creation process performed by the dimension creation device according to Embodiment 2;

FIG. 22 is a diagram of an example 3D model for which dimensions are created with a dimension creation device according to Embodiment 3;

FIG. 23 is a diagram of the 3D model in FIG. 22 in a 2D plane;

FIG. 24 is a functional block diagram of a dimension creation device according to Embodiment 4;

FIG. 25 is a diagram of an example 2D drawing in Embodiment 4; and

FIG. 26 is a flowchart of a process of creating a 2D drawing in Embodiment 4.

DESCRIPTION OF EMBODIMENTS

Embodiments of the present disclosure will be described below in detail with reference to the drawings. The same reference signs denote the same or corresponding components in the drawings.

Embodiment 1

FIG. 1 is a schematic diagram of a dimension creation device 10 according to Embodiment 1. The dimension creation device 10 is a three-dimensional (3D) CAD device and is also an information processing device that assists in designing an object having a solid shape using a computer. Examples of an object to be designed include a product in the manufacturing process and an element of the product.

The dimension creation device 10 assists in creating a dimension between two elements included in elements of a 3D model. The 3D model herein refers to a model representing a solid shape of an object to be designed. The elements of the 3D model indicate faces, edges, and vertices forming a solid shape of a 3D model. The dimension between two elements indicates the distance, or more specifically, the length between two elements.

For example, FIG. 2 shows a 3D model M1. FIG. 3 shows the shape of the 3D model M1 in a two-dimensional (2D) plane when viewed in three directions including the X-direction, the Y-direction, and the Z-direction. As shown in FIGS. 2 and 3, the 3D model M1 represents a hexahedron having six faces S1 to S6, twelve edges R1 to R12, and eight vertices V1 to V8 as elements. More specifically, when viewed in the X-direction, the 3D model M1 is rectangular. When viewed in the Y-direction, the 3D model M1 is trapezoidal, with the face S2 being a slope.

The dimension creation device 10 creates dimensional information indicating the dimension between two elements selected by an operator from multiple faces, multiple edges,

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and multiple vertices included in the 3D model M1, and annotates the 3D model M1 with the created dimensional information.

The dimensional information created for the 3D model M1 by the dimension creation device 10 is read with software including computer aided manufacturing (CAM) or computer aided testing (CAT), and used for automatically creating a processing program or a measurement program. For example, a computer numeric control (CNC) 3D measuring device can automatically create a measurement program. More specifically, the measuring device can specify, from the dimensions provided to the 3D model M1, two faces associated with the dimensions, and automatically generate an operation path of the measuring device that performs 3D measurement of the faces. The measuring device can also automatically create a program for comparing the results of the 3D measurement with the dimensions.

To correctly read and process the dimensions created by the dimension creation device 10 with such subsequent software, the dimensions created for the 3D model M1 are to be correctly associated with elements of the 3D model M1.

Referring back to FIG. 1, the dimension creation device 10 includes a controller 11, a storage 12, an operation receiver 13, a display 14, and a communicator 15.

The controller 11 includes a central processing unit (CPU), a read-only memory (ROM), and a random-access memory (RAM). The CPU may also be referred to as a central processing device, a central arithmetic device, a processor, a microprocessor, a microcomputer, or a digital signal processor (DSP), and functions as a central arithmetic processing unit that performs processing and arithmetic operations for controlling the dimension creation device 10. In the controller 11, the CPU reads programs and data stored in the ROM, and centrally controls the dimension creation device 10 using the RAM as a work area.

The storage 12 includes a nonvolatile semiconductor memory such as a flash memory, an erasable programmable ROM (EPROM), or an electrically erasable programmable ROM (EEPROM) to serve as a secondary storage or an auxiliary storage. The storage 12 stores programs and data used by the controller 11 to perform various processes. The storage 12 also stores data generated or acquired by the controller 11 through various processes. For example, the storage 12 stores the 3D model M1 read by the dimension creation device 10 and dimensional information created by the dimension creation device 10.

The operation receiver 13 includes input devices such as a keyboard, a mouse, a switch, a touchpad, and a touchscreen to receive operations on the dimension creation device 10 from an operator. For example, the operation receiver 13 receives an operation of reading the 3D model M1, operations of zooming in and out and rotating the 3D model M1 displayed on the display 14, and an operation of selecting any of elements of the 3D model M1.

The display 14 includes a display device such as a liquid crystal display (LCD) panel or an organic electro-luminescence (EL) panel. The display 14 is driven by a display driving circuit (not shown), and displays various images under control of the controller 11. For example, the display 14 displays the 3D model M1 shown in FIG. 2.

The communicator 15 includes a communication interface to communicate with devices external to the dimension creation device 10. For example, the communicator 15 communicates with external devices in accordance with a

known communication protocol such as a wireless local area network (LAN), a wired LAN, or a universal serial bus (USB).

The functional configuration of the dimension creation device **10** will now be described with reference to FIG. 4.

As shown in FIG. 4, the dimension creation device **10** functionally includes a placement plane indicator **110**, an element selector **120**, a direction indicator **130**, a dimension creator **140**, and an output device **150**. The dimension creator **140** functions as an association-element determiner **141** and an associator **142**. These functions are implemented by software, firmware, or a combination of software and firmware. The software and the firmware are described as programs and stored in the ROM or the storage **12**. When the CPU executes the programs stored in the ROM or the storage **12**, these functions are implemented.

In the example described below, the dimension creation device **10** creates dimensions for the 3D model **M1** shown in FIG. 2.

The placement plane indicator **110** indicates a dimension placement plane. The dimension placement plane is a plane in which the dimensions created by the dimension creation device **10** are placed. FIG. 5 shows an example dimension placement plane **P1** with a dot-and-dash line. The dimension placement plane **P1** is an imaginary plane parallel to the face **S1** of the 3D model **M1** and at a predetermined distance from the face **S1**.

The placement plane indicator **110** indicates, for example, the plane shown in FIG. 5 as the dimension placement plane **P1** in accordance with an operation received from the operator through the operation receiver **13**. The operator operates the operation receiver **13** to indicate any plane in the 3D space as the dimension placement plane **P1**. More specifically, the plane parallel to the direction of the dimension to be created is indicated as the dimension placement plane **P1** to place the dimension at an easily-visible position. The placement plane indicator **110** is implemented by the controller **11** operating in cooperation with the operation receiver **13**. The placement plane indicator **110** functions as placement-plane indication means.

Referring back to FIG. 4, the element selector **120** selects two target elements from the elements of the 3D model. The two target elements are herein used to create a target dimension and define the two ends of the dimension.

For example, with the display **14** displaying the 3D model **M1**, the operator operates the operation receiver **13**, and clicks intended two of the vertices, edges, and faces of the multiple elements included in the 3D model **M1**. The element selector **120** selects the two target elements in accordance with the operation received from the operator through the operation receiver **13**. The element selector **120** is implemented by the controller **11** operating in cooperation with the operation receiver **13**. The element selector **120** functions as element selection means.

More specifically, when selecting the two target elements, the operator operates the operation receiver **13** to change the orientation of the 3D model **M1** displayed on the display **14** to align with the orientations of the screen and the dimension placement plane **P1** indicated by the placement plane indicator **110**. More specifically, as shown in FIG. 6, the operator rotates the 3D model **M1** displayed on the display **14** to align the orientation of the dimension placement plane **P1** with the orientation of the screen. Thus, the 3D model **M1** is viewed in the Z-direction that is orthogonal to the dimension placement plane **P1**.

This adjustment of the orientation of the 3D model **M1** enables creation of the dimensions for the 3D model **M1**

with the same operability as for creating the dimensions for a 2D drawing. Thus, the operator can easily select two target elements at the two ends of a target dimension.

However, such an operation of changing the orientation of the 3D model **M1** may not be performed. Two target elements can be selected by selecting individual elements while, for example, the 3D model **M1** is being rotated, without the orientation of the dimension placement plane **P1** being aligned with the orientation of the screen.

In the example described below, two dimensions including a dimension **L1** corresponding to the length of the edges **R2**, **R4**, **R6**, and **R8**, and a dimension **L2** corresponding to the length of the edges **R1** and **R3** are created.

To create the dimension **L1**, the element selector **120** selects, in accordance with the operation of the operator, target elements from vertices, edges, or faces at the two ends of the dimension **L1**. For example, the element selector **120** selects edges **R1** and **R3** indicated with bold lines in FIG. 7 as target elements. In another example, the element selector **120** selects, for example, vertices **V1** and **V3** indicated with large dots in FIG. 8 as target elements.

To create the dimension **L2**, the element selector **120** selects, in accordance with the operation of the operator, target elements from vertices, edges, or faces located at the two ends of the dimension **L2**. For example, the element selector **120** selects edges **R2** and **R4** indicated with bold lines in FIG. 9 as target elements. In another example, the element selector **120** selects, for example, the vertices **V1** and **V3** indicated with large dots in FIG. 8 as target elements.

Referring back to FIG. 4, the direction indicator **130** indicates the direction of the target dimension. The operator operates the operation receiver **13** to indicate any direction in a 3D coordinate system including the X-direction, the Y-direction, and the Z-direction. For example, to create the dimension **L1**, the direction of the dimension **L1** corresponds to the Y-direction, and thus the operator indicates the Y-direction in the 3D coordinate system. In another example, to create the dimension **L2**, the direction of the dimension **L2** corresponds to the X-direction, and thus the operator indicates the X-direction in the 3D coordinate system.

As described above, the direction indicator **130** indicates the direction of the dimension in accordance with the operation received from the operator through the operation receiver **13**. The direction indicator **130** is implemented by the controller **11** operating in cooperation with the operation receiver **13**. The direction indicator **130** functions as direction indication means.

In addition to the direction of the dimension, the direction indicator **130** also indicates the position at which the dimension is placed. The operator operates the operation receiver **13** and clicks an intended position in the dimension placement plane **P1** indicated by the placement plane indicator **110** to indicate the position at which the dimension is placed. The direction indicator **130** thus indicates a position at which the dimension created by the dimension creation device **10** is placed in accordance with the operation received from the operator through the operation receiver **13**.

The dimension creator **140** creates the dimension between the two target elements selected by the element selector **120** in the direction indicated by the direction indicator **130**. More specifically, the dimension creator **140** determines whether the 3D model **M1** includes a first orthogonal face and a second orthogonal face when the element selector **120** selects vertices or edges as a first target element and a second target element. The first orthogonal face includes the

selected first target element, and is orthogonal to the direction indicated by the direction indicator 130. The second orthogonal face includes the selected second target element, and is orthogonal to the direction indicated by the direction indicator 130. When the 3D model M1 is determined to include both the first orthogonal face and the second orthogonal face, the dimension creator 140 creates the dimension between the first orthogonal face and the second orthogonal face in the direction indicated by the direction indicator 130 in a manner associated with the first orthogonal face and the second orthogonal face.

When the dimension is indicated in a drawing, the target elements indicated by the dimension are normally expected to be faces, edges, or vertices with a higher likelihood in the stated order. Thus, the dimension is associated with faces irrespective of vertices or edges selected as the target elements. More specifically, when, for example, a target element of any dimension can be expected to be either a face or an edge, the indication of the dimension is expected to be an indication of the face. For example, when the dimension L1 in FIG. 6 is indicated, the indication can be expected to be an indication of a distance between two parallel faces that are the faces S5 and S6, a distance between the edges R1 and R3, or a distance between the edges R5 and R7. In this case, the expectation to be faces is more likely, and thus the indication is expected to be the distance between the faces S5 and S6. The distance between the edges R1 and R3 and the distance between the edges R5 and R7 are the same as the distance between the faces S5 and S6, and thus can be uniquely determined when the distance between the faces S5 and S6 is determined.

In an inspection process, the distance between the faces S5 and S6 is measured in accordance with the above expectation. Unless indicated otherwise, the distance between the edges R1 and R3 and the distance between the edges R5 and R7 are not measured. For example, irrespective of the edges R1 and R3 selected as the target elements to create the dimension L1, the element that is to be indicated by a designer is the distance between the faces S5 and S6, and the elements that are to be measured for the dimension L1 in the inspection process are the faces S5 and S6. When elements to be associated with the indicated dimension is unchanged from the edges R1 and R3 selected as the target elements without the faces S5 and S6 being associated with the dimension, the intended faces S5 and S6 are not measured in a measurement program created with the dimensional information read by software with CAT. Thus, the faces S5 and S6 are associated with the dimension irrespective of the edges R1 and R3 being selected.

As described above, irrespective of vertices or edges selected as target elements, any faces that can be allocated as target elements of the dimension are to be determined as target elements. Thus, when faces have the selected vertices or edges and are orthogonal to the direction of the dimension, the dimension creator 140 associates the dimension with the faces, instead of the selected vertices or edges.

More specifically, the dimension creator 140 creates the dimension associated with the association elements with the association-element determiner 141 and the associator 142.

The association-element determiner 141 determines two association elements based on the two target elements selected by the element selector 120 and the direction indicated by the direction indicator 130. The association elements are elements associated with the target dimension among the elements of the 3D model M1. The association-element determiner 141 determines the elements appropriate for defining the target dimension among faces, edges, and

vertices in the elements of the 3D model M1 as the association elements. The association-element determiner 141 functions as association-element determination means.

(1) In a first case described below, the element selector 120 selects edges as two target elements.

When the element selector 120 selects an edge as the first target element, the association-element determiner 141 determines whether the 3D model M1 includes a first orthogonal face including the edge selected by the element selector 120 and orthogonal to the direction indicated by the direction indicator 130. Similarly, when the element selector 120 selects an edge as the second target element, the association-element determiner 141 determines whether the 3D model M1 includes a second orthogonal face including the edge selected by the element selector 120 and orthogonal to the direction indicated by the direction indicator 130.

For example, to create the dimension L1 as shown in FIG. 7, the edge R1 may be selected as the first target element, and the edge R3 may be selected as the second target element. In this case, the faces including the edge R1 among the six faces S1 to S6 of the 3D model are the faces S1 and S5 colored in FIG. 10. Of the faces S1 and S5, the face orthogonal to the Y-direction that is the direction of the dimension L1 is the face S5. Thus, the association-element determiner 141 determines that the 3D model M1 has the face S5 as the first orthogonal face.

The faces including the edge R3 among the six faces S1 to S6 of the 3D model are the face S6 hatched in FIG. 10 and the face S1. Of the faces S1 and S6, the face orthogonal to the Y-direction that is the direction of the dimension L1 is the face S6. Thus, the association-element determiner 141 determines that the 3D model M1 has the face S6 as the second orthogonal face.

When the 3D model M1 is determined as including the face S5 as the first orthogonal face, the association-element determiner 141 determines the face S5 as a first association element. Similarly, when the 3D model M1 is determined as including the face S6 as the second orthogonal face, the association-element determiner 141 determines the face S6 as the second association element. Thus, the association-element determiner 141 determines the face S5 colored in FIG. 11 as the first association element and the face S6 colored in FIG. 11 as the second association element.

In contrast, when none of the faces including the edge selected by the element selector 120 are orthogonal to the direction of the dimension, no face is appropriate for defining the target dimension. In other words, when no face is orthogonal to the direction of the dimension, the dimension is not expected to be a distance indicated for the faces. Thus, when the 3D model M1 does not include the first orthogonal face, the association-element determiner 141 determines the edge serving as the first target element and selected by the element selector 120 as the first association element. Similarly, when the 3D model M1 does not include the second orthogonal face, the association-element determiner 141 determines the edge serving as the second target element and selected by the element selector 120 as the second association element.

For example, to create the dimension L2 as shown in FIG. 9, the edge R2 may be selected as the first target element, and the edge R4 may be selected as the second target element. In this case, the faces including the edge R2 among the six faces S1 to S6 of the 3D model are the faces S1 and S4 colored in FIG. 12. Of the faces S1 and S4, the face that is orthogonal to the X-direction that is the direction of the dimension L2 is the face S4. Thus, the association-element determiner 141 determines that the 3D model M1 has the

face **S4** as the first orthogonal face, and determines the face **S4** as the first association element.

The faces including the edge **R4** among the six faces **S1** to **S6** of the 3D model are the face **S2** hatched in FIG. 12 and the face **S1**. The faces **S1** and **S2** are not orthogonal to the X-direction that is the direction of the dimension **L2**. Thus, the association-element determiner **141** determines that the 3D model **M1** does not include the second orthogonal face.

When the 3D model **M1** is determined as not including the second orthogonal face, the association-element determiner **141** determines the edge **R4** selected as the second target element as the second association element instead of a face. Thus, the association-element determiner **141** determines the face **S4** colored in FIG. 13 as the first association element and the edge **R4** indicated with a bold solid line as the second association element.

(2) In a second case described below, the element selector **120** selects vertices as two target elements.

When the element selector **120** selects a vertex as the first target element, the association-element determiner **141** determines whether the 3D model **M1** includes a first orthogonal edge including the vertex selected by the element selector **120** and orthogonal to the direction indicated by the direction indicator **130**. Similarly, when the element selector **120** selects a vertex as the second target element, the association-element determiner **141** determines whether the 3D model **M1** includes a second orthogonal edge including the vertex selected by the element selector **120** and orthogonal to the direction indicated by the direction indicator **130**.

More specifically, to create the dimension **L2** as shown in FIG. 8, the vertex **V1** may be selected as the first target element, and the vertex **V3** may be selected as the second target element. In this case, the edges including the vertex **V1** among the twelve edges **R1** to **R12** of the 3D model are the edges **R1**, **R2**, and **R9** indicated with bold solid lines in FIG. 14. Of the edges **R1**, **R2**, and **R9**, the edges orthogonal to the X-direction that is the direction of the dimension **L2** are the edges **R2** and **R9**. Thus, the association-element determiner **141** determines that the 3D model **M1** includes the edges **R2** and **R9** as the first orthogonal edges.

The edges including the vertex **V3** among the twelve edges **R1** to **R12** of the 3D model are the edges **R3**, **R4**, and **R11** indicated with bold broken lines in FIG. 14. Of the edges **R3**, **R4**, and **R11**, the edge orthogonal to the X-direction that is the direction of the dimension **L2** is the edge **R4**. Thus, the association-element determiner **141** determines that the 3D model **M1** includes the edge **R4** as the second orthogonal edge.

When the 3D model **M1** includes any first orthogonal edge, the association-element determiner **141** further determines whether the 3D model **M1** includes the first orthogonal face including the first orthogonal edge and orthogonal to the direction indicated by the direction indicator **130**. Similarly, when the 3D model **M1** includes any second orthogonal edge, the association-element determiner **141** further determines whether the 3D model **M1** includes the second orthogonal face including the second orthogonal edge and orthogonal to the direction indicated by the direction indicator **130**.

In the example shown in FIG. 14, the face orthogonal to the direction of the dimension **L2** among the faces including the edge **R2** that is one of the first orthogonal edges is the face **S4**. In addition, the face orthogonal to the direction of the dimension **L1** among the faces including the edge **R9** that is another one of the first orthogonal edges is also the face **S4**. Thus, the association-element determiner **141** determines that the 3D model **M1** has the face **S4** as the first

orthogonal face. When the 3D model **M1** has the face **S4** as the first orthogonal face, the association-element determiner **141** determines the face **S4** as the first association element.

In contrast, the faces including the edge **R4** that is one of the second orthogonal edges are the faces **S1** and **S2**. None of these faces **S1** and **S2** are orthogonal to the direction of the dimension **L2**, indicating that no face is appropriate for defining the target dimension. When the 3D model **M1** does not include the second orthogonal face, the association-element determiner **141** determines the edge **R4** serving as the second orthogonal edge as the second association element.

When the vertices **V1** and **V3** are selected as the target elements to create the dimension **L2**, the association-element determiner **141** determines the face **S4** colored in FIG. 13 and the edge **R4** indicated with a bold solid line as the two association elements.

When none of the edges including the vertex selected by the element selector **120** are orthogonal to the direction of the dimension (not shown), no edge is appropriate for defining the target dimension. When the 3D model **M1** does not include any first orthogonal edge, the association-element determiner **141** determines the vertex selected by the element selector **120** as the first association element. Similarly, when the 3D model **M1** does not include any second orthogonal edge, the association-element determiner **141** determines the vertex selected by the element selector **120** as the second association element.

As described above, when the element selector **120** selects a vertex or an edge as the first target element and the 3D model **M1** has a face including the first target element and orthogonal to the direction of the dimension indicated by the direction indicator **130**, the association-element determiner **141** determines the face as the first association element. Similarly, when the element selector **120** selects a vertex or an edge as the second target element and the 3D model **M1** has a face including the second target element and orthogonal to the direction of the dimension indicated by the direction indicator **130**, the association-element determiner **141** determines the face as the second association element. In addition, when the 3D model **M1** does not include the first orthogonal face and the second orthogonal face, the association-element determiner **141** further determines edges or vertices as the first association element and the second association element. Thus, the association-element determiner **141** automatically determines the association elements depending on the expectation of the target dimension to be the distance between faces, edges, or vertices.

In addition to a vertex or an edge, the element selector **120** can also select the first target element or the second target element from the multiple faces of the 3D model. In other words, the operator can directly select a face orthogonal to the direction of the dimension as at least one of the first target element or the second target element. When the element selector **120** selects a face orthogonal to the direction of the dimension as the first target element or the second target element, the association-element determiner **141** determines the selected face as the first association element or the second association element.

Referring back to FIG. 4, the associator **142** associates the target dimension with the first association element and the second association element determined by the association-element determiner **141**. More specifically, the associator **142** associates the dimension **L1** with the faces **S5** and **S6** to create the dimension **L1**. In another example, the associator **142** associates the dimension **L2** with the face **S4** and the edge **R4** to create the dimension **L2**.

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Thus, the associator **142** creates dimensional information indicating a target dimension and two association elements associated with the dimension. The created dimensional information defines that the dimension between two target elements selected by the element selector **120** is the dimension between the two association elements determined by the association-element determiner **141**. The associator **142** functions as association means.

In the manner described above, the dimension creator **140** creates the dimension associated with the two association elements determined by the association-element determiner **141** as the dimension between the two target elements selected by the element selector **120** in the direction indicated by the direction indicator **130**. The dimension creator **140** creates the dimension associated with the two association elements at the position indicated by the direction indicator **130** in the dimension placement plane **P1** indicated by the placement plane indicator **110**. The dimension creator **140** is implemented by the controller **11**. The dimension creator **140** functions as dimension creation means.

Referring back to FIG. **4**, the output device **150** outputs, outside the dimension creation device **10**, the dimensional information indicating the dimension created by the dimension creator **140**. For example, the output device **150** causes the display **14** to display the 3D model **M1** annotated with the dimension created by the dimension creator **140**. As shown in, for example, FIG. **11** or FIG. **13**, the output device **150** causes the two association elements associated with the dimension **L1** or **L2** to appear in an emphasized manner to be visible for the operator.

In another example, the output device **150** outputs, to an external device through the communicator **15**, output information indicating the 3D model **M1** annotated with the dimension created by the dimension creator **140** and indicating the two association elements associated with the dimension. The external device is a device for, for example, processing or measuring a product based on the dimensional information created by the dimension creation device **10**. The output device **150** is implemented by the controller **11** operating in cooperation with the display **14** or the communicator **15**. The output device **150** functions as output means.

With reference to FIG. **15**, the procedure of a dimension creation process performed by the dimension creation device **10** will be described. The dimension creation process shown in FIG. **15** is started after the 3D model **M1** for which a dimension is created is read by the dimension creation device **10**, and the 3D model **M1** is displayed on the display **14**.

When the dimension creation process is started, the controller **11** first functions as the placement plane indicator **110**, and indicates the dimension placement plane **P1** (step **S11**). More specifically, the controller **11** receives the operation of the operator through the operation receiver **13**, and indicates, for example, the dimension placement plane **P1** shown in FIG. **5** in accordance with the received operation.

The controller **11** then functions as the element selector **120**, and selects two target elements for creating the dimension (step **S12**). More specifically, the controller **11** receives the operation of the operator through the operation receiver **13**, and selects vertices or edges at the two ends of the target dimension in accordance with the received operation.

The controller **11** then functions as the direction indicator **130** and indicates the direction and the position of the dimension (step **S13**). More specifically, the controller **11** receives an operation of the operator through the operation receiver **13**. The controller **11** indicates an appropriate direction in the 3D space as the direction of the target

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dimension in accordance with the operation received from the operator, and indicates an appropriate position in the dimension placement plane **P1** as a position at which the target dimension is placed.

After indicating or selecting the dimension placement plane **P1**, the two target elements, and the direction and the position of the dimension, the controller **11** functions as the association-element determiner **141** and determines the first association element corresponding to the first target element of the two target elements selected in step **S12** (step **S14**). The details of the process in step **S14** will be described with reference to the flowchart shown in FIG. **16**.

Upon starting an association-element determination process shown in FIG. **16**, the controller **11** sets a variable **Et** indicating the first association element to the first target element selected in step **S12** (step **S101**).

After setting the variable **Et**, the controller **11** determines whether the variable **Et** is a vertex (step **S102**). In other words, the controller **11** determines whether the first target element selected in step **S12** is a vertex.

When the variable **Et** is a vertex (YES in step **S102**), the controller **11** determines whether any edge including the vertex is orthogonal to the direction of the dimension indicated in step **S13** (step **S103**). In other words, when the first target element selected in step **S12** is a vertex, the controller **11** determines whether the 3D model **M1** includes at least one first orthogonal edge including the selected vertex and orthogonal to the direction of the indicated dimension.

When the first orthogonal edge is included (YES in step **S103**), the controller **11** sets the variable **Et** to the edge (step **S104**). When, for example, the first target element selected in step **S12** is the vertex **V1**, the edges including the vertex **V1** and orthogonal to the direction of the dimension are the edges **R1** and **R9**. Thus, the controller **11** sets the variable **Et** to the edges **R1** and **R9**.

In contrast, when the first orthogonal edge is not included (NO in step **S103**), the controller **11** ends the association-element determination process shown in FIG. **16**. In this case, the variable **Et** indicating the first association element is set to the vertex selected in step **S12**. Thus, the controller **11** determines the vertex selected in step **S12** as the first association element.

When the first target element selected in step **S12** is not a vertex, or more specifically, when the first target element is an edge or a face, the controller **11** determines that the variable **Et** is not a vertex in step **S102** (NO in step **S102**). In this case, the controller **11** skips the processes in steps **S103** and **S104**.

When the variable **Et** is set to an edge in step **S104** or when the variable **Et** is determined as not being a vertex in step **S102**, the controller **11** advances the processing to step **S105**. In step **S105**, the controller **11** determines whether the variable **Et** is an edge (step **S105**). When the variable **Et** is an edge, an edge is selected in step **S12**, or a vertex is selected in step **S12** and any of the edges including the vertex is orthogonal to the direction of the dimension.

When the variable **Et** is an edge (YES in step **S105**), the controller **11** determines whether any of the faces including the edge is orthogonal to the direction of the dimension indicated in step **S13** (step **S106**). In other words, when the first target element selected in step **S12** is an edge or an edge is set to the variable **Et** in step **S104**, the controller **11** determines whether the 3D model **M1** includes the first orthogonal face including the edge and orthogonal to the indicated direction.

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When the first orthogonal face is included (YES in step S106), the controller 11 sets the variable Et to the face (step S107). When, for example, the first target element selected in step S12 is the edge R1, the face including the edge R1 and orthogonal to the direction of the dimension corresponds to the face S5. Thus, the controller 11 sets the variable Et to the face S5.

In another example, when the first target element selected in step S12 is the vertex V1 and the variable Et is set to the edges R2 and R9 in step S104, the face including the edge R2 and orthogonal to the direction of the dimension and the face including the edge R9 and orthogonal to the direction of the dimension each correspond to the face S5. Thus, the controller 11 sets the variable Et to the face S5.

In contrast, when the first orthogonal face is not included (NO in step S106), the controller 11 ends the association-element determination process shown in FIG. 16. In this case, the variable Et indicating the first association element is set to the edge selected in step S12 or set in step S104. Thus, the controller 11 determines the edge as the first association element.

When the first target element selected in step S12 is a vertex, and none of the edges including the vertex are orthogonal to the direction of the dimension, the controller 11 determines that the variable Et is not an edge in step S105 (NO in step S105). In another example, when the first target element selected in step S12 is neither a vertex nor an edge, or more specifically, when the first target element is a face, the controller 11 determines that the variable Et is not an edge in step S105. In these cases, the controller 11 skips the processes in steps S106 and S107. The association-element determination process in step S14 then ends.

Referring back to FIG. 15, after determining the first association element corresponding to the first target element, the controller 11 determines the second association element corresponding to the second target element of the two target elements selected in step S12 (step S15). The process in step S15 is the same as the process in step S14 described with reference to FIG. 16. More specifically, the process in step S15 can be similarly described by replacing the first target element in the process in step S14 with the second target element, the first association element with the second association element, the first orthogonal face with the second orthogonal face, and the first orthogonal edge with the second orthogonal edge.

After determining the two association elements corresponding to the two target elements in this manner, the controller 11 functions as the associator 142, and creates the dimension (step S16). More specifically, the controller 11 associates the dimension between the two association elements determined in steps S14 and S15 in the direction indicated in step S13 with these two association elements. Thus, the controller 11 creates dimensional information indicating the target dimension and the two association elements associated with the dimension.

After creating the dimension, the controller 11 functions as the output device 150, and outputs the created dimension (step S17). More specifically, the controller 11 causes the display 14 to display the 3D model M1 for which the dimension is created. In another example, the controller 11 generates output information indicating the 3D model M1 for which the dimension is created and the two association elements associated with the dimension, and outputs the generated information to an external device through the communicator 15. The dimension creation process shown in FIG. 15 then ends.

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As described above, the dimension creation device 10 according to Embodiment 1 selects the first target element and the second target element from the elements of the 3D model M1, and indicates the direction of the target dimension. When vertices or edges are selected as the first target element and the second target element, and the 3D model M1 includes the first orthogonal face including the first target element and orthogonal to the direction of the dimension and the second orthogonal face including the second target element and orthogonal to the direction of the dimension, the dimension creation device 10 according to Embodiment 1 creates the dimension between the first orthogonal face and the second orthogonal face in the indicated direction in a manner associated with the first orthogonal face and the second orthogonal face.

As described above, irrespective of the operator selecting a vertex or an edge, the dimension is created in a manner associated with any face to be associated with the dimension. The operator thus does not determine the element to be associated with the dimension. This allows efficient creation of a dimension correctly associated with an element of the 3D model M1 created efficiently. The dimension creation device 10 according to Embodiment 1 can create a dimension correctly readable with software such as CAM or CAT, and thus can enhance the design work efficiency using a 3D model.

A dimension associated with a face can be created irrespective of the operator selecting a vertex or an edge. Thus, the two faces at the two ends of the target dimension are not to be selected directly. This structure can reduce the operation of rotating the 3D model M1 to display a face not appearing at the front of the screen. This enables selection of elements of a 3D model with the same operability as in selecting elements of a 2D model, and can enhance the work efficiency of the operator.

Embodiment 2

Embodiment 2 will now be described. The same components and functions as in Embodiment 1 will not be described.

A dimension creation device 10 according to Embodiment 2 has functions of the placement plane indicator 110, the element selector 120, the direction indicator 130, the dimension creator 140, and the output device 150 described in Embodiment 1. The dimension creator 140 in Embodiment 2 has, in addition to the functions described in Embodiment 1, a function of determining whether the 3D model includes an element with the same dimension as the target dimension other than the two association elements determined by the association-element determiner 141.

FIG. 17 shows a 3D model M2 for describing the functions of the dimension creation device 10 according to Embodiment 2. The 3D model M2 has the shape of the 3D model M1 described in Embodiment 1 with the face S5 partly being cutout. More specifically, the face S5 in the 3D model M1 is divided into faces S25 and S27 in the same plane in the 3D model M2. For simplicity, in FIG. 17 and the subsequent figures, some of the faces, the edges, and the vertices included in the 3D model M2 are not denoted with the reference signs.

In Embodiment 2, the association-element determiner 141 determines whether the 3D model M2 includes at least one element different from the first association element and the second association element in an imaginary plane including at least one of the first association element or the second association element and orthogonal to the direction of the

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dimension indicated by the direction indicator **130**. When the 3D model **M2** is determined as including the at least one element, the associator **142** further associates the target dimension with the at least one element in addition to the first association element and the second association element.

In the example described below, as shown in FIG. **18**, the element selector **120** selects vertices **V21** and **V23** as the two target elements to create a dimension **L3** corresponding to the length of edge **R22**, **R24**, or **R28** of the 3D model **M2**. FIG. **18** shows a dimension placement plane **P2** indicated by the placement plane indicator **110** with a dot-and-dash line.

When the element selector **120** selects the vertex **V21** as the first target element, the association-element determiner **141** determines the edge **R21** including the vertex **V21** and orthogonal to the direction of the dimension **L3** as the first orthogonal edge. The association-element determiner **141** then determines the face **S25** including the edge **R21** serving as the first orthogonal edge and orthogonal to the direction of the dimension **L3** as the first orthogonal face.

Similarly, when the element selector **120** selects the vertex **V23** as the second target element, the association-element determiner **141** determines the edges **R23** and **R31** including the vertex **V23** and orthogonal to the direction of the dimension **L3** as the second orthogonal edges. The association-element determiner **141** then determines the face **S26** including the edges **R23** and **R31** serving as the second orthogonal edges and orthogonal to the direction of the dimension **L3** as the second orthogonal face.

Thus, the association-element determiner **141** determines the face **S25** colored in FIG. **19** as the first association element and the face **S26** colored in FIG. **19** as the second association element. The associator **142** associates the target dimension **L3** with the faces **S25** and **S26** determined as the first and the second association elements by the association-element determiner **141**.

In Embodiment 2, when the faces **S25** and **S26** are determined as the first and the second association elements in this manner, the association-element determiner **141** determines whether the 3D model **M2** includes at least one face different from the faces **S25** and **S26** in an imaginary plane including at least one of the faces **S25** and **S26** and orthogonal to the direction of the dimension **L3**. In other words, the association-element determiner **141** determines whether at least one face different from the faces **S25** and **S26** is in the same plane as at least one of the faces **S25** and **S26**. When the 3D model **M2** includes at least one face different from the faces **S25** and **S26** in the same plane, the association-element determiner **141** determines the at least one face as an additional association element.

More specifically, the 3D model **M2** includes a face **S27** colored in FIG. **20** as another face in the same plane as the face **S25** serving as the first orthogonal face. The faces **S25** and **S27** are in the same plane. Thus, the dimension between the faces **S27** and **S26** is the dimension **L3** in the same manner as for the dimension between the faces **S25** and **S26**. Thus, the association-element determiner **141** determines the face **S27** as an additional association element.

In contrast, the 3D model **M2** includes none of other faces, edges, and vertices in the same plane as the face **S26** serving as the second orthogonal face. Thus, the association-element determiner **141** determines that the same plane as the face **S26** includes no additional association element.

When the association-element determiner **141** determines the face **S27** as the additional association element, the associator **142** further associates the target dimension **L3** with the face **S27** in addition to the faces **S25** and **S26**. The dimension creator **140** thus creates the dimension associated

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with the three faces **S25**, **S26**, and **S27**. In other words, the dimension creator **140** creates the dimension **L3** as the dimension between the faces **S25** and **S26** and between the faces **S26** and **S27**. Thus, when the 3D model **M2** is read with software such as CAM or CAT, in addition to the faces **S25** and **S26**, the face **S27** can be also acquired as a target element from the dimension **L3**. When, for example, the dimension **L3** is selected in CAT, in addition to the faces **S25** and **S26**, the face **S27** can be also set as a face to be measured. To acquire dimensional information from faces, information about the dimension **L3** can be acquired from the face **S27** in addition to the faces **S25** and **S26**.

With reference to FIG. **21**, the procedure of a dimension creation process performed by the dimension creation device **10** according to Embodiment 2 will now be described. The processes in steps **S21** to **S25** in the dimension creation process shown in FIG. **21** are the same as the processes in steps **S11** to **S15** in the dimension creation process shown in FIG. **15** in Embodiment 1, and thus will not be described.

The controller **11** performs the processes in steps **S21** to **S25** to determine the first association element and the second association element associated with the target dimension. After determining the first association element and the second association element, the controller **11** determines whether at least one element different from the first association element and the second association element is included as an element of the 3D model **M2** in an imaginary plane including at least one of the first association element or the second association element and orthogonal to the direction of the dimension (step **S26**).

When the at least one element is included (YES in step **S26**), the controller **11** determines the at least one different element as an additional association element associated with the target dimension (step **S27**). When the at least one different element is not included (NO in step **S26**), the controller **11** skips the process in step **S27**.

After determining the association element as above, the controller **11** creates the dimension in a manner associated with the at least two association elements determined in steps **S24**, **S25**, and **S27** (step **S28**). More specifically, when determining the additional association element in step **S27**, the controller **11** creates the dimension in a manner associated with the additional association element and the two association elements determined in steps **S25** and **26**. When the additional association element is not determined in step **S27**, the controller **11** creates the dimension in a manner associated with the two association elements determined in steps **S25** and **26**.

After creating the dimension, the controller **11** outputs the created dimension (step **S29**). The process in step **S29** is the same as the process in step **S17** in Embodiment 1. The dimension creation process shown in FIG. **21** then ends.

As described above, when the 3D model **M2** includes at least one different element in an imaginary plane including at least one of the first association element or the second association element and orthogonal to the direction of the dimension, the dimension creation device **10** according to Embodiment 2 creates a dimension in a manner associated with the at least one different element in addition to the first association element and the second association element. Thus, another element with the same dimension as the target dimension is automatically detected and associated with the dimension. Thus, time and effort for creating dimensions for multiple elements can be reduced. The operator can efficiently add elements to be associated with the dimension.

In Embodiment 2, the first association element and the second association element are faces. However, the first

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association element or the second association element that are a vertex or an edge instead of a face can be similarly described. More specifically, when the first association element or the second association element is a vertex or an edge, the association-element determiner **141** determines whether the 3D model **M2** includes at least one different element in an imaginary plane including the vertex or the edge and orthogonal to the direction of the dimension. When the 3D model **M2** includes at least one different element, the associator **142** further associates the target dimension with the at least one different element in addition to the first association element and the second association element.

The at least one different element may be a vertex or an edge instead of a face. More specifically, when the 3D model **M2** includes an edge without including a face as the at least one different element, the association-element determiner **141** determines the edge as the additional association element. When the 3D model **M2** includes a vertex without including any face or any edge as the at least one different element, the association-element determiner **141** determines the vertex as the additional association element. Thus, irrespective of the different element with the same dimension the target dimension being a vertex or an edge, the operator can reduce time and effort taken to create the dimension.

Embodiment 3

Embodiment 3 will now be described. The same components and functions as in Embodiments 1 and 2 will not be described.

In Embodiments 1 and 2, the edge selected as the target element is a straight edge. In Embodiment 3 described below, a circular edge is included in elements of the 3D model and selected as a target element. In this case, instead of being a perfect circle, the circle may have a shape slightly different from a perfect circle such as an ellipse.

FIG. 22 shows a 3D model **M3** in Embodiment 3. FIG. 23 shows the shape of the 3D model **M3** when viewed in the positive Z-direction in a 2D plane. As shown in FIGS. 22 and 23, as in the 3D model **M1** described in Embodiment 1, the 3D model **M3** includes six faces **S1** to **S6**, twelve edges **R1** to **R12**, and eight vertices **V1** to **V8** as elements. The 3D model **M3** additionally has two cylindrical through-holes extending in the Z-direction. The 3D model **M3** has two cylindrical faces **S10** and **S11** serving as the surfaces defining the two through-holes.

In the example described below, a dimension **L4** is created for the 3D model **M3**. The dimension **L4** corresponds to the distance between the face **S4** and a cylinder axis **A1** of the cylindrical face **S10**. To be distinguished from the cylindrical faces **S10** and **S11**, the face **S4** is referred to as a flat face **S4** below.

The element selector **120** selects two target elements for creating the dimension **L4** in accordance with the operation of the operator. More specifically, the element selector **120** selects the edge **R2** as the first target element and the edge **R13** as the second target element. The edge **R2** is a straight edge included in the flat face **S4**. In contrast, the edge **R13** is a circular edge included in the cylindrical face **S10**. The operator operates the operation receiver **13** to select, from the elements in the 3D model **M3**, the edges **R2** and **R13** as the two target elements for creating the dimension **L4**.

The direction indicator **130** indicates the X-direction as a direction of the dimension **L4** in accordance with the operation of the operator. The direction indicator **130** also indi-

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cates, in accordance with the operation of the operator, a position in the dimension placement plane **P1** at which the dimension **L4** is placed.

When the straight edge **R2** is selected as the first target element, the association-element determiner **141** determines the first association element by performing the same determination process as in Embodiment 1.

More specifically, the association-element determiner **141** determines whether the 3D model **M3** includes a first orthogonal face including the edge **R2** selected by the element selector **120** and orthogonal to the direction indicated by the direction indicator **130**. The 3D model **M3** includes the flat face **S4** as the first orthogonal face, and thus the association-element determiner **141** determines the flat face **S4** as the first association element.

In contrast, when the circular edge **R13** is selected as the second target element, the association-element determiner **141** determines whether the 3D model **M3** includes a second orthogonal cylindrical face including the circular edge **R13** and a cylinder axis orthogonal to the direction indicated by the direction indicator **130**.

In the 3D model **M3**, the circular edge **R13** is one of the two circular edges included in the cylindrical face **S10**. The cylinder axis **A1** of the cylindrical face **S10** extends in the Z-direction and is orthogonal to the X-direction that is the direction of the dimension **L4**. Thus, the association-element determiner **141** determines that the 3D model **M3** includes the cylindrical face **S10** as the second orthogonal cylindrical face. When the circular edge **R13** is selected as the second target element and the 3D model **M3** includes a second orthogonal cylindrical face as above, the association-element determiner **141** determines the cylindrical face **S10** serving as the second orthogonal cylindrical face as the second association element.

When the association-element determiner **141** determines the first association element and the second association element, the associator **142** associates the target dimension **L4** with the flat face **S4** serving as the first association element and determined by the association-element determiner **141** and with the cylindrical face **S10** serving as the second association element. Thus, the dimension creator **140** creates the dimension **L4** associated with the flat face **S4** and the cylindrical face **S10**. The output device **150** outputs, outside the dimension creation device **10**, dimensional information indicating the dimension **L4** created by the dimension creator **140**. The processes performed by the associator **142** and the output device **150** are the same as those in Embodiment 1, and will not be described in detail.

In contrast, when the element selector **120** selects a circular edge as the second target element, and the 3D model **M3** does not include the second orthogonal cylindrical face (not shown), the association-element determiner **141** determines the circular edge selected as the second target element as the second association element. In other words, when none of the cylindrical faces included in the 3D model **M3** include a circular edge selected as the second target element and a cylinder axis orthogonal to the direction indicated by the direction indicator **130**, the 3D model **M3** includes no face appropriate for defining the target dimension. In this case, the association-element determiner **141** determines a circular edge that is a second target element selected by the element selector **120** as the second association element.

Although the circular edge **R13** is selected as the second target element in Embodiment 3, the process is similarly performed as when a circular edge is selected as the first target element. More specifically, the same process is performed for the first target element and the second target

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element. Thus, the process for the circular edge as the first target element can be described similarly for the second target element replaced with the first target element, the second orthogonal cylindrical face with the first orthogonal cylindrical face, and the second association element with the first association element in the example described above.

As described above, when a circular edge is selected as the target element, and the 3D model includes a cylindrical face including the circular edge and a cylinder axis orthogonal to the direction of the target dimension, the dimension creation device **10** according to Embodiment 3 creates the target dimension in a manner associated with the cylindrical face. Irrespective of a circular edge selected as the target element, the dimension creation device **10** according to Embodiment 3 creates the dimension in a manner associated with any cylindrical face appropriate for association with the dimension. Thus, the operator does not determine the element to be associated with the dimension. Thus, a dimension correctly associated with the element of the 3D model can be created efficiently.

Irrespective of a circular edge selected as a target element, the dimension associated with the cylindrical face can be created. Thus, the operator does not directly select the cylindrical face. This structure can reduce the operation of the operator rotating the 3D model to display a cylindrical face not displayed at the front of the screen. This enables selection of elements of a 3D model with the same operability as in selecting elements of a 2D model, and can thus enhance the work efficiency of the operator.

In Embodiment 3, when at least one of the first association element or the second association element is a cylindrical face, and an imaginary plane including the cylinder axis of the cylindrical face and orthogonal to the direction indicated by the direction indicator **130** includes the cylinder axis of at least one cylindrical face different from the first association element and the second association element, the dimension creator **140** creates a dimension by further associating the dimension with the at least one cylindrical face in addition to the first association element and the second association element.

More specifically, when the at least one of the first association element or the second association element is a cylindrical face, the association-element determiner **141** determines whether an imaginary plane including the cylinder axis of the cylindrical face and orthogonal to the direction of the dimension indicated by the direction indicator **130** includes the cylinder axis of a cylindrical face different from the first association element and the second association element. When the imaginary plane is determined as including the cylinder axis of the at least one cylindrical face, the associator **142** further associates the target dimension with the at least one cylindrical face in addition to the first association element and the second association element.

In the example described below, similarly to the above, the element selector **120** selects the flat face **S4** as the first association element and the cylindrical face **S10** as the second association element to create the dimension **L4**. In this case, an imaginary plane including a cylinder axis **A1** of the cylindrical face **S10** serving as the second association element and orthogonal to the X-direction that is the direction of the dimension **L4** includes a cylinder axis **A2** of a cylindrical face **S11**. Thus, the association-element determiner **141** determines the cylindrical face **S11** as the additional association element.

When the association-element determiner **141** determines the cylindrical face **S11** as the additional association ele-

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ment, the associator **142** further associates the target dimension **L4** with the cylindrical face **S11** in addition to the flat face **S4** and the cylindrical face **S10**. The dimension creator **140** in this manner creates the dimension **L4** associated with the flat face **S4** and the cylindrical faces **S10** and **S11**.

As described above, when the first association element or the second association element is a cylindrical face, and an imaginary plane including the cylinder axis of the cylindrical face and orthogonal to the direction of the target dimension includes the cylinder axis of a different cylindrical face, the dimension creation device **10** according to Embodiment 3 creates the dimension by further associating the dimension with the different cylindrical face. As described above, another element with the same dimension as the target dimension is automatically detected and associated with the dimension. Thus, time and effort for creating dimensions for multiple elements can be reduced. The operator can efficiently add elements to be associated with the dimension.

Embodiment 4

Embodiment 4 will now be described. The same components and functions as in Embodiments 1 to 3 will not be described.

In Embodiments 1 to 3, the element selector **120** selects the first target element and the second target element for creating a dimension from faces, edges, and vertices serving as elements of the 3D model in accordance with the operation of the operator. The direction indicator **130** indicates an appropriate direction in the 3D coordinate system as the direction of the dimension in accordance with the operation of the operator. In Embodiment 4, the element selector **120** selects the first target element and the second target element based on a 2D drawing representing a 3D model in a 2D plane. The direction indicator **130** indicates the direction of the dimension based on the 2D drawing.

FIG. **24** is a functional block diagram of a dimension creation device **10a** according to Embodiment 4. The dimension creation device **10a** according to Embodiment 4 functionally includes a placement plane indicator **110**, an element selector **120**, a direction indicator **130**, a dimension creator **140**, an output device **150**, and a 2D drawing creator **160**. These functions are implemented by software, firmware, or a combination of software and firmware. The software and the firmware are described as programs, and the programs are stored in the ROM or the storage **12**. When the CPU executes the programs stored in the ROM or the storage **12**, these functions are implemented.

The 2D drawing creator **160** creates, from a 3D model, a 2D drawing corresponding to the 3D model. The 2D drawing represents the 3D model in a 2D plane by projecting the 3D model in the 2D plane with a 3D CAD function.

For example, FIG. **25** shows a 2D drawing **D1** of the 3D model **M1** shown in FIG. **2**. The 2D drawing **D1** is created by projecting the 3D model **M1** in the 2D plane. More specifically, the 2D drawing **D1** includes a projection chart **Q1** acquired by projecting the 3D model **M1** in the positive Z-direction, with the positive Y-direction in a 3D coordinate system being upward.

The projection chart **Q1** shows the shape of the 3D model **M1** in the positive Z-direction, or more specifically, when viewed from the faces **S1** and **S2**. The projection chart **Q1** includes, among the elements of the 3D model **M1**, elements corresponding to the edges **R1** to **R4**, **R8**, **R11**, and **R12** and the vertices **V1** to **V4**, **V7**, and **V8** viewable in the positive Z-direction. For example, edges **TR1** and **TR3** in the projection chart **Q1** correspond to the edges **R1** and **R3** in the

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3D model M1, and vertices TV1 and TV3 in the projection chart Q1 correspond to the vertices V1 and V3 in the 3D model M1.

In addition to the projection chart Q1, the 2D drawing D1 includes information indicating two dimensions TL1 and TL2. The dimensions TL1 and TL2 are dimensions in a 2D coordinate system including the 2D drawing D1. More specifically, the dimension TL1 is the vertical dimension between the edges TR1 and TR3. The dimension TL2 is the lateral dimension between the vertices TV1 and TV3.

With reference to FIG. 26, a process of creating a 2D drawing with the 2D drawing creator 160 will now be described. The 2D drawing creator 160 starts the process shown in FIG. 26 upon receiving, from the operator through the operation receiver 13, an instruction of creating a 2D drawing from a 3D model.

After starting the process of creating a 2D drawing, the 2D drawing creator 160 indicates a projection direction of the 3D model in accordance with the operation of the operator (step S41). For example, to create the 2D drawing D1 shown in FIG. 25 from the 3D model M1 shown in FIG. 2, the operator operates the operation receiver 13 to indicate the positive Z-direction as the projection direction. In another example, to create a 2D drawing by projecting the front face of a 3D model, the operator operates the operation receiver 13 to indicate the direction perpendicular to the front face as the projection direction. In still another example, to create a 2D drawing by projecting a side face of the 3D model, the operator operates the operation receiver 13 to indicate the direction perpendicular to the side face as the projection direction.

Subsequently, the 2D drawing creator 160 indicates a projection position of the 3D model in accordance with the operation of the operator (step S42). The operator operates the operation receiver 13 to indicate any position at which the 2D drawing is created in an area in the 2D plane in which the 3D model is projected.

After indicating the projection direction and the projection position, the 2D drawing creator 160 projects the 3D model in the indicated projection direction and at the projection position (step S43). When, for example, the 3D model M1 shown in FIG. 2 is projected in the positive Z-direction, the projection chart Q1 as shown in FIG. 25 is created as the resultant shape of the 3D model M1 being projected in the 2D plane.

After creating the projection chart, the 2D drawing creator 160 selects target elements for creating a dimension in the 2D drawing in accordance with the operation of the operator, and indicates the direction (step S44). For example, to create the dimension TL1, the operator operates the operation receiver 13 to select the edges TR1 and TR3 as target elements from edges and vertices serving as the elements of the projection chart Q1 and indicate the vertical direction as the direction of the dimension. To create the dimension TL2, the operator operates the operation receiver 13 to select the vertices TV1 and TV3 as target elements from edges and vertices serving as the elements of the projection chart Q1 and indicate the lateral direction as the direction of the dimension.

After selecting the target elements and indicating the direction, the 2D drawing creator 160 creates the dimension in the created 2D drawing (step S45). When, for example, the edges TR1 and TR3 are indicated as the target elements, and the lateral direction is indicated as the direction of the dimension, the 2D drawing creator 160 creates the dimension TL1 in a manner associated with the edges TR1 and TR3. When the vertices TV1 and TV3 are indicated as the

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target elements, and the vertical direction is indicated as the direction of the dimension, the 2D drawing creator 160 creates the dimension TL2 in a manner associated with the vertices TV1 and TV3.

With the above processes, the 2D drawing creator 160 creates, from a 3D model, a 2D drawing representing the 3D model in a 2D plane. The 2D drawing creator 160 is implemented by the controller 11 operating in cooperation with the operation receiver 13. The 2D drawing creator 160 functions as 2D drawing creation means.

Edges and vertices serving as elements of the 2D drawing are associated with the elements of the 3D model. Thus, when the shape or the dimension of the 3D model changes, the shape or the dimension of the 2D drawing changes accordingly. When, for example, the dimension of the edge R1 of the 3D model M1 changes, the dimension of the edge TR1 corresponding to the edge R1 in the 2D drawing D1 changes accordingly. When the position of the vertex V1 of the 3D model M1 changes, the position of the vertex TV1 corresponding to the vertex V1 in the 2D drawing D1 changes accordingly.

In the manner described above, the dimension creation device 10a according to Embodiment 4 creates the dimension for the 3D model based on the 2D drawing created by the 2D drawing creator 160. As described above, when the 2D drawing D1 including the two dimensions TL1 and TL2 is created by the 2D drawing creator 160, the dimension creation device 10a creates dimensions corresponding to the two dimensions TL1 and TL2 for the 3D model M1.

A process performed for the dimension TL1 will first be described. The element selector 120 selects the first target element and the second target element from the elements of the 3D model M1 based on the edges TR1 and TR3 associated with the dimension TL1 in the 2D drawing D1. More specifically, the element selector 120 specifies the edges TR1 and TR3 associated with the dimension TL1 from edges and vertices serving as the elements of the 2D drawing D1. The edges TR1 and TR3 are target elements of the dimension TL1 selected by the operator upon creation of the dimension TL1.

After specifying the edges TR1 and TR3, the element selector 120 specifies the edges R1 and R3 corresponding to the specified edges TR1 and TR3 from the elements of the 3D model M1. The element selector 120 then selects the specified edges R1 and R3 as the first target element and the second target element.

The element selector 120 may determine either one of the edges R1 and R3 as the first target element or as the second target element. For example, the element nearer the origin of the coordinate may be determined as the first target element in accordance with a rule.

The direction indicator 130 indicates the direction of the target dimension in the coordinate system of the 3D model M1 based on the direction of the dimension TL1 in the 2D drawing D1. The direction indicator 130 first specifies the vertical direction that is the direction of the dimension TL1 in the 2D drawing D1. The direction indicator 130 then calculates the direction corresponding to the direction of the dimension TL1 in the coordinate system of the 3D model M1 based on the projection direction of the 3D model M1 upon creation of the 2D drawing D1.

More specifically, as described above, the projection chart Q1 included in the 2D drawing D1 is created by projecting the 3D model M1 in the positive Z-direction, with the positive Y-direction being upward. Thus, the vertical direction in the 2D drawing D1 corresponds to the Y-direction in the coordinate system of the 3D model M1.

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In this manner, the direction indicator **130** specifies the directional correspondence between the coordinate system of the 2D drawing **D1** and the coordinate system of the 3D model **M1** from the projection direction of the 3D model **M1** upon creation of the 2D drawing **D1**. The direction indicator **130** then calculates, based on the directional correspondence, the direction corresponding to the direction of the dimension **TL1** in the 3D coordinate system as the Y-direction. The direction indicator **130** indicates the calculated Y-direction as the direction of the dimension created for the 3D model **M1**.

After the element selector **120** selects the edges **R1** and **R3** as the two target elements and the direction indicator **130** indicates the Y-direction as the direction of the dimension, the dimension creator **140** creates the dimension in the indicated Y-direction between the selected edges **R1** and **R3**. The process performed by the dimension creator **140** is the same as the process described in Embodiment 1, and is not described. In the manner as in FIG. 11, the dimension **L1** associated with the faces **S5** and **S6** is finally created for the 3D model **M1**.

A process performed for the dimension **TL2** will now be described. The element selector **120** selects the first target element and the second target element from the elements of the 3D model **M1** based on the vertices **TV1** and **TV3** associated with the dimension **TL2** in the 2D drawing **D1**. More specifically, the element selector **120** specifies the vertices **TV1** and **TV3** associated with the dimension **TL2** from edges and vertices serving as the elements of the 2D drawing **D1**. The vertices **TV1** and **TV3** are target elements of the dimension **TL2** selected by the operator upon creation of the dimension **TL2**.

After specifying the vertices **TV1** and **TV3**, the element selector **120** specifies the vertices **V1** and **V3** corresponding to the specified vertices **TV1** and **TV3** from the elements of the 3D model **M1**. The element selector **120** selects the specified vertices **V1** and **V3** as the first target element and the second target element.

The element selector **120** may determine either one of the vertices **V1** and **V3** as the first target element or as the second target element. For example, the element nearer the origin of the coordinate may be determined as the first target element in accordance with a rule.

The direction indicator **130** indicates the direction of the target dimension in the coordinate system of the 3D model **M1** based on the direction of the dimension **TL2** in the 2D drawing **D1**. The direction indicator **130** first specifies the lateral direction that is the direction of the dimension **TL2** in the 2D drawing **D1**. The direction indicator **130** then calculates the direction corresponding to the direction of the dimension **TL2** in the coordinate system of the 3D model **M1** based on the projection direction of the 3D model **M1** upon creation of the 2D drawing **D1**.

More specifically, the lateral direction that is the direction of the dimension **TL2** in the 2D drawing **D1** corresponds to the X-direction in the coordinate system of the 3D model **M1**. Thus, the direction indicator **130** calculates, based on this directional correspondence, the direction corresponding to the direction of the dimension **TL2** in the 3D coordinate system as the X-direction. The direction indicator **130** indicates the calculated X-direction as the direction of the dimension created for the 3D model **M1**.

After the element selector **120** selects the vertices **V1** and **V3** as the two target elements and the direction indicator **130** indicates the X-direction as the direction of the dimension as above, the dimension creator **140** creates the dimension in the indicated X-direction between the selected vertices **V1**

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and **V3**. The process performed by the dimension creator **140** is the same as the process described in Embodiment 1, and is not described. In the same manner as in FIG. 13, the dimension **L2** associated with the face **S4** and the edge **R4** is finally created for the 3D model **M1**.

As described above, the dimension creation device **10a** according to Embodiment 4 selects the first target element and the second target element and indicates the direction of the target dimension based on the 2D drawing representing the 3D model in the 2D plane. When a 3D model created in the past and a 2D drawing created by projecting the 3D model are available, a dimension correctly associated with elements of the 3D model can be automatically created without the operator selecting target elements from the 3D model or indicating the direction. This structure can enhance the work efficiency.

In the example described above, the 2D drawing **D1** includes the projection chart **Q1** acquired by projecting the 3D model **M1** in the positive Z-direction. However, the 2D drawing may include multiple projection charts acquired by projecting the 3D model in multiple directions, and a dimension may be created for each of the multiple projection charts. For a dimension created for each of the multiple projection charts, the dimension creation device **10a** can create the dimensions corresponding to all the dimensions included in the 2D drawing for the 3D model by performing the processes similar to the above processes on all the dimensions.

In the example described above, the dimension creation device **10a** functions as the 2D drawing creator **160**. However, the dimension creation device **10a** may not function as the 2D drawing creator **160**. In other words, the dimension creation device **10a** may not create a 2D drawing from a 3D model, but may acquire a 2D drawing created by another device having a 3D CAD function to perform the above process on the acquired 2D drawing.

In the example described above, the 2D drawing is created by projecting a 3D model. Thus, the elements of the 2D drawing and the elements of the 3D model are associated with each other upon creation of the 3D model. However, a 2D drawing representing a 3D model in a 2D plane may be created in any manner other than being created by projecting the 3D model. For example, a 2D drawing may be printed matter or an image representing the shape of a 3D model viewed in a specific direction.

When the 2D drawing is created in a manner other than projecting a 3D model, the element selector **120** extracts a dimension from the 2D drawing through an image analysis of the 2D drawing, and specifies a line in the 2D drawing to which the dimensional line of the dimension is connected. The element selector **120** then matches the shapes of the 2D drawing and the 3D model with each other to specify the elements of the 3D model corresponding to the extracted line in the 2D drawing. The direction indicator **130** matches the shapes of the 2D drawing and the 3D model with each other to specify the directional correspondence between the coordinate system of the 2D drawing and the coordinate system of the 3D model. The direction indicator **130** then calculates the direction in the coordinate system of the 3D model corresponding to the direction of the dimension extracted from the 2D drawing. As described above, irrespective of the elements of the 2D drawing and the elements of the 3D model not associated with each other upon creation of the 3D model, the dimension creation device **10a** can create a dimension associated with the elements of the 3D model.

The embodiments of the present disclosure described above may be modified or applied in various forms to implement the present disclosure.

For example, in the above embodiments, the functions of the dimension creation devices **10** and **10a** are described using the 3D models **M1** to **M3**. However, the 3D models **M1** to **M3** are mere examples. A dimension may be created for a 3D model having another solid shape in the manner described in the above embodiments.

In the above embodiments, the association-element determiner **141** determines whether the 3D model **M1** includes a first orthogonal face or a second orthogonal face by determining whether any face is orthogonal to the direction indicated by the direction indicator **130**. However, when a plane parallel to the direction of the dimension is indicated as the dimension placement plane **P1** or **P2**, being orthogonal to the direction indicated by the direction indicator **130** corresponds to being perpendicular to the dimension placement plane **P1** or **P2**. Thus, the association-element determiner **141** may determine whether the 3D model **M1** includes any face orthogonal to the direction indicated by the direction indicator **130** by determining whether the 3D model **M1** has a face perpendicular to the dimension placement plane **P1** or **P2**.

In the above embodiments, the dimension creation device **10** or **10a** includes the operation receiver **13** and the display **14**. However, the dimension creation device **10** or **10a** may eliminate the operation receiver **13** and the display **14**. In other words, the dimension creation device **10** or **10a** may not have the function of a user interface. In this case, the dimension creation device **10** or **10a** receives an operation from the operator through an external input device and displays an image on the external display device.

In the above embodiments, the CPU in the controller **11** in the dimension creation device **10** or **10a** executes a program stored in the ROM or the storage **12** to function as each unit shown in FIG. **4** or **24**. However, the controller **11** may be dedicated hardware. Examples of the dedicated hardware include a single circuit, a complex circuit, a programmed processor, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or a combination of two or more of these. When the controller **11** is dedicated hardware, the function of each unit may be implemented by individual hardware devices, or the functions of the units may be collectively implemented by a single hardware device.

Some functions of the unit may be implemented by dedicated hardware, and others may be implemented by software or firmware. The controller **11** can thus implement the above functions with hardware, software, firmware, or a combination of two or more of these.

When a program defining the operation of the dimension creation device **10** or **10a** according to one or more embodiments of the present disclosure is applied to an existing computer such as a personal computer or an information terminal, the computer can function as the dimension creation device **10** or **10a** according to one or more embodiments of the present disclosure.

The above program may be distributed in any method. For example, the program may be stored in a non-transitory computer-readable recording such as a compact disk ROM (CD-ROM), a digital versatile disk (DVD), a magneto-optical disk (MO), or a memory card for distribution or may be distributed through a communication network such as the Internet.

The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full range of equivalents to which such claims are entitled.

This application claims the benefit of Japanese Patent Application No. 2020-022114, filed on Feb. 13, 2020, the entire disclosure of which is incorporated by reference herein.

REFERENCE SIGNS LIST

10, 10a Dimension creation device
11 Controller
12 Storage
13 Operation receiver
14 Display
15 Communicator
110 Placement plane indicator
120 Element selector
130 Direction indicator
140 Dimension creator
141 Association-element determiner
142 Associator
150 Output device
160 Two-dimensional (2D) drawing creator
A1, A2 Cylinder axis
D1 Two-dimensional (2D) drawing
L1 to L4, TL1, TL2 Dimension
M1 to M3 3D model
P1, P2 Dimension placement plane
Q1 Projection chart
R1 to R14, R21 to R32, TR1, TR3 Edge
S1 to S6, S21 to S27 Face
S10, S11 Cylindrical face
V1 to V8, V21 to V28, TV1, TV3 Vertex

The invention claimed is:

1. A dimension creation device, comprising:
 element selecting circuitry to receive selection of a first target element and a second target element from elements of a three-dimensional model in a first viewing plane, the three-dimensional (3D) model including an arrangement of plural faces, plural edges, and plural vertices, wherein each of the selected first target element and the selected second target element includes a vertex of the plural vertices or an edge of the plural edges;
 direction indicating circuitry to receive a direction of a dimension to be created in the first viewing plane; and
 dimension creating circuitry including:
 association-element determining circuitry that identifies two elements of the 3D model that are associated with the dimension to be created, the association-element determining circuitry:
 determining whether the first target element is part of a first orthogonal face of the plural faces by identifying each of the plural faces that includes the first target element and determining whether one of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created;

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determining whether the second target element is part of a second orthogonal face of the plural faces by identifying each of the plural faces that includes the second target element and determining whether one of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created; and

identifying one of the vertex, the edge, the first orthogonal face, or the second orthogonal face as a first association element or a second association element based on the determinations of the first target element and the second target element, and the direction of the dimension to be created, wherein:

when the association-element determining circuitry determines that one of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created, the association element determining circuitry identifies the first orthogonal face as the first association element,

when the association-element determining circuitry determines that one of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created, the association element determining circuitry identifies the second orthogonal face as the second association element,

when none of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created, the association element determining circuitry identifies the vertex or the edge selected as the first target element as the first association element, and

when none of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created, the association element determining circuitry identifies the vertex or the edge selected as the second target element as the second association element; and

associating circuitry to associate the dimension to be created with the first association element and the second association element that are identified by the association-element determining circuitry,

wherein the dimension creating circuitry creates the dimension in the first viewing plane between the first target element and the second target element in the direction indicated by the direction indicating circuitry based on the first association element and the second association element identified by the association-element determining circuitry.

2. The dimension creation device according to claim 1, wherein

the association-element determining circuitry determines, when the element selecting circuitry selects a vertex as the first target element, whether the three-dimensional model includes a first orthogonal edge including the vertex selected by the element selecting circuitry and orthogonal to the direction indicated by the direction indicating circuitry, and determines the vertex selected by the element selecting circuitry as the first association element when the three-dimensional model does not include the first orthogonal edge, and

the association-element determining circuitry determines, when the element selecting circuitry selects a vertex as the second target element, whether the three-dimensional

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sional model includes a second orthogonal edge including the vertex selected by the element selecting circuitry and orthogonal to the direction indicated by the direction indicating circuitry, and determines the vertex selected by the element selecting circuitry as the second association element when the three-dimensional model does not include the second orthogonal edge.

3. The dimension creation device according to claim 2, wherein

the association-element determining circuitry further determines, when the three-dimensional model includes the first orthogonal edge, whether the three-dimensional model includes the first orthogonal face, determines the first orthogonal face as the first association element when the three-dimensional model includes the first orthogonal face, and determines the first orthogonal edge as the first association element when the three-dimensional model does not include the first orthogonal face, and

the association-element determining circuitry further determines, when the three-dimensional model includes the second orthogonal edge, whether the three-dimensional model includes the second orthogonal face, determines the second orthogonal face as the second association element when the three-dimensional model includes the second orthogonal face, and determines the second orthogonal edge as the second association element when the three-dimensional model does not include the second orthogonal face.

4. The dimension creation device according to claim 1, wherein

the association-element determining circuitry determines, when the element selecting circuitry selects a circular edge as the first target element and the three-dimensional model includes a first orthogonal cylindrical face including the circular edge and a cylinder axis orthogonal to the direction indicated by the direction indicating circuitry, the first orthogonal cylindrical face as a first association element, and

the association-element determining circuitry determines, when the element selecting circuitry selects a circular edge as the second target element and the three-dimensional model includes a second orthogonal cylindrical face including the circular edge and a cylinder axis orthogonal to the direction indicated by the direction indicating circuitry, the second orthogonal cylindrical face as a second association element.

5. The dimension creation device according to claim 4, wherein

the association-element determining circuitry determines, when the element selecting circuitry selects a circular edge as the first target element and the three-dimensional model does not include the first orthogonal cylindrical face, the circular edge as the first association element, and

the association-element determining circuitry determines, when the element selecting circuitry selects a circular edge as the second target element and the three-dimensional model does not include the second orthogonal cylindrical face, the circular edge as the second association element.

6. The dimension creation device according to claim 1, wherein

when the three-dimensional model includes at least one different element different from the first association element and the second association element in an imaginary plane including at least one of the first

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association element or the second association element and orthogonal to the direction indicated by the direction indicating circuitry, the dimension creating circuitry creates the dimension in a manner further associated with the at least one different element in addition to the first association element and the second association element.

7. The dimension creation device according to claim 1, wherein

when at least one of the first association element or the second association element is a cylindrical face, and an imaginary plane including a cylinder axis of the cylindrical face and orthogonal to the direction indicated by the direction indicating circuitry includes a cylinder axis of at least one different cylindrical face different from the first association element and the second association element, the dimension creating circuitry creates the dimension in a manner further associated with the at least one different cylindrical face in addition to the first association element and the second association element.

8. The dimension creation device according to claim 1, further comprising:

placement-plane indicating circuitry to indicate a dimension placement plane in which the dimension is placed, wherein the dimension creating circuitry creates the dimension in the dimension placement plane indicated by the placement-plane indicating circuitry.

9. The dimension creation device according to claim 1, wherein

the direction indicating circuitry further indicates a position at which the dimension is placed, and the dimension creating circuitry creates the dimension at the position indicated by the direction indicating circuitry.

10. The dimension creation device according to claim 1, wherein

the element selecting circuitry selects the first target element and the second target element based on a two-dimensional drawing representing the three-dimensional model in a two-dimensional plane, and the direction indicating circuitry indicates the direction of the dimension to be created based on the two-dimensional drawing.

11. The dimension creation device according to claim 10, wherein

the element selecting circuitry specifies an edge or a vertex associated with a dimension created in the two-dimensional drawing from elements of the two-dimensional drawing, and selects, from the elements of the three-dimensional model, an edge or a vertex corresponding to the specified edge or the specified vertex as each of the first target element and the second target element, and

the direction indicating circuitry specifies a direction of the dimension created in the two-dimensional drawing, and indicates, in a coordinate system of the three-dimensional model, a direction corresponding to the specified direction as the direction of the dimension to be created.

12. A dimension creation method, comprising:

selecting, by a controller of a computing device, a first target element and a second target element from elements of a three-dimensional model in a first viewing plane, the three-dimensional model including an arrangement of plural faces, plural edges, and plural vertices, wherein each of the selected first target ele-

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ment and the selected second target element includes a vertex of the plural vertices or an edge of the plural edges;

receiving, by an operation receiver of the computing device, a direction of a dimension to be created in the first viewing plane;

identifying, by the controller of the computing device, two elements of the 3D model that are associated with the dimension to be created;

determining, by the controller of the computing device, whether the first target element is part of a first orthogonal face of the plural faces by identifying each of the plural faces that includes the first target element and determining whether one of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created;

determining, by the controller of the computing device, whether the second target element is part of a second orthogonal face of the plural faces by identifying each of the plural faces that includes the second target element and determining whether one of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created; and

identifying, by the controller of the computing device, one of the vertex, the edge, the first orthogonal face, or the second orthogonal face as a first association element or a second association element based on the determinations of the first target element and the second target element, and the direction of the dimension to be created, wherein:

when one of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created, identifying the first orthogonal face as the first association element,

when one of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created, identifying the second orthogonal face as the second association element,

when none of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created, identifying the vertex or the edge selected as the first target element as the first association element, and

when none of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created, identifying the vertex or the edge selected as the second target element as the second association element;

associating, by the controller of the computing device, the dimension to be created with the first association element and the second association element; and

creating, by the controller of the computing device, the dimension in the first viewing plane between the first target element and the second target element in a specified direction based on the first association element and the second association element.

13. A non-transitory computer-readable recording medium storing a program, the program causing a computer to:

receive selection of a first target element and a second target element from elements of a three-dimensional model in a first viewing plane, the three-dimensional model including an arrangement of plural faces, plural edges, and plural vertices, wherein each of the selected

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first target element and the selected second target element includes a vertex of the plural vertices or an edge of the plural edges,
 receive a direction of a dimension to be created in the first viewing plane, and
 identify two elements of the 3D model that are associated with the dimension to be created;
 determine whether the first target element is part of a first orthogonal face of the plural faces by identifying each of the plural faces that includes the first target element and determining whether one of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created;
 determine whether the second target element is part of a second orthogonal face of the plural faces by identifying each of the plural faces that includes the second target element and determining whether one of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created; and
 identify one of the vertex, the edge, the first orthogonal face, or the second orthogonal face as a first association element or a second association element based on the determinations of the first target element and the second target element, and the direction of the dimension to be created, wherein:

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when one of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created, identify the first orthogonal face as the first association element,
 when one of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created, identify the second orthogonal face as the second association element,
 when none of the plural faces that includes the first target element is orthogonal to the direction of the dimension to be created, identify the vertex or the edge selected as the first target element as the first association element, and
 when none of the plural faces that includes the second target element is orthogonal to the direction of the dimension to be created, identify the vertex or the edge selected as the second target element as the second association element;
 associate the dimension to be created with the first association element and the second association element;
 and
 create the dimension in the first viewing plane between the first target element and the second target element in a specified direction based on the first association element and the second association element.

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